

Predicting Energy Assimilation from Resource Quantity and Quality: A Mechanistic DEB Approach for *Aporrectodea caliginosa*

Supplementary information

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1 Earthworm culture

1.1 Breeding and experiment soil

The soil used in the culture of earthworms is collected from the top 10 cm of a meadow in Versailles called “Les Closeaux”. It is then dried before being milled at < 2 mm.

Table 1.1: Main characteristics of the breeding soil from Versailles. Table from Bart et al. (2017).

Characteristics	Value
Clay (< 2 μm , g/kg)	226.0
Fine silt (2-20 μm , g/kg)	174.1
Coarse silt (20-50 μm , g/kg)	298.9
Fine sand (50-200 μm , g/kg)	239.1
Coarse sand (200-2000 μm , g/kg)	47.9
CaCO_3 total (g/kg)	23.3
Organic matter (g/kg)	32.6
P_2O_5 (g/kg)	0.1
Organic carbon (g/kg)	18.9
Total nitrogen (N) (g/kg)	1.5
C/N	12.5
pH	7.5
Total Cu (mg/kg)	25.2

1.2 DNA barcoding

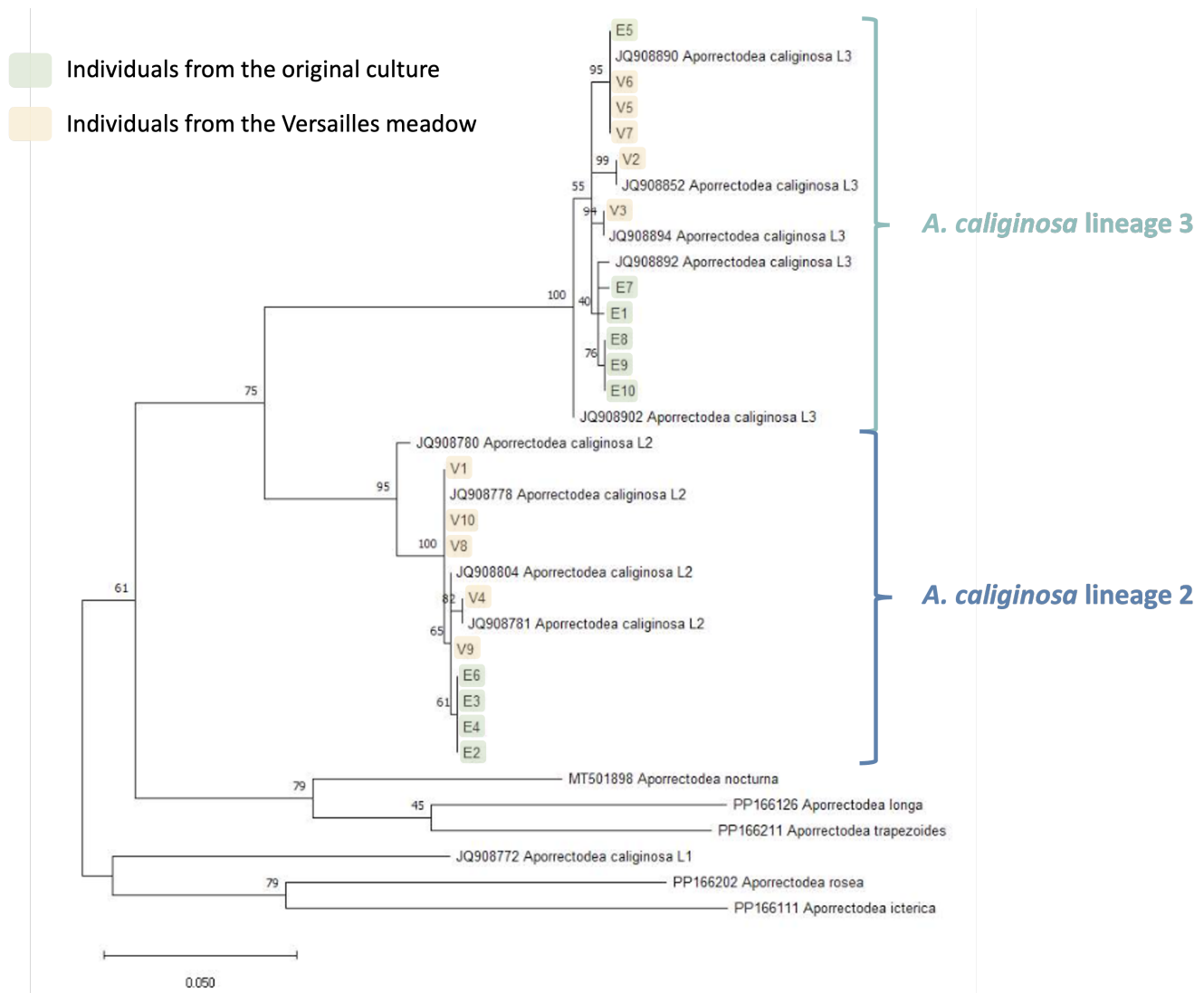


Figure 1.1: Results of the DNA barcoding of 10 individuals from the culture (highlighted in yellow)

The phylogenetic tree was realised thanks to maximum likelihood bootstrap method (1000 simulations) with the Tamura-Nei model (nucleotides). Bootstrap frequency is indicated on each branch.

Aporrectodea caliginosa is a cryptic species complex comprising three lineages. Our earthworm culture is known to include individuals from two closely related lineages (2 and 3). While these lineages could potentially exhibit differences in sensitivity, previous research on *Gammarus roeselii* has shown that tolerance within cryptic species complexes is not always lineage dependent (Kabus et al., 2024).

2 Models

2.1 Model formulation

2.1.1 Model state and intermediate variables

Table 2.1: Model state and intermediate variables.

Symbol	Description	Unit
Environmental state variables		
OM_{horse}	Horse dung organic matter mass	g_{OM}
OM_{soil}	Soil organic matter mass	g_{OM}
$Soil_{w,cosm}$	Dry mass of soil in the cosm	g_{soil}
Organism state variables		
E	Energy reserve	J
E_h	Maturity	J
L	Volumetric structural length	cm
R	Cumulated reproduction	#
W_w	Wet weight	g
Intermediate variables		
E_m	$\frac{\{\dot{p}_{Am}\}}{v}$	$J.cm^{-1}$
f	Scaled functional response	—
g	$\frac{[E_G]}{\kappa_{real}E_m}$	cm^{-2}
K_m	$\frac{[\dot{p}_M]_t}{[E_G]}$	d^{-1}
κ_{real}	Allocation fraction to soma	—
L_m	$\frac{v_t}{K_m g}$	cm
$\dot{OM}_{out}^{horse}$	Rate of consumption of horse dung organic matter	$g_{OM}.d^{-1}$
$\dot{OM}_{out}^{soil}$	Rate of consumption of soil organic matter	$g_{OM}.d^{-1}$
$\dot{p}_C$	Flux of mobilized energy from reserve	$J.d^{-1}$

Symbol	Description	Unit
$\dot{p}_R$	Energy flux allocated to maturity / reproduction	J.d ⁻¹
$\dot{Q}_{f,max}$	Maximum rate of energy ingested	J.d ⁻¹
$\dot{Q}_f^{matter}$	Rate of matter ingestion	J.d ⁻¹
$\dot{Q}_f^{horse}$	Rate of energy ingested through horse dung	J.d ⁻¹
$\dot{Q}_f^{soil}$	Rate of energy ingested through soil	J.d ⁻¹

2.1.2 Standard DEB model

Reserve energy dynamic

$$\frac{dE}{dt} = \frac{\{\dot{p}_{Am}\}_t}{L} \left(f - \frac{E \nu_t}{\{\dot{p}_{Am}\}_t} \right)$$

with :

- E : Energy in the reserve (J)
- ν_t : Energy conductance at temperature T_{exp} ($cm.d^{-1}$)

Growth

$$\frac{dL}{dt} = \frac{\nu_t}{3 \left(\frac{E}{E_m} + g \right)} \left(\frac{E}{E_m} - \frac{L}{L_m} \right)$$

$$K_m = \frac{[\dot{p}_{M,t}]}{[E_G]}$$

$$E_m = \frac{\{\dot{p}_{Am}\}_t}{\nu_t}$$

$$g = \frac{[E_G]}{\kappa E_m}$$

$$L_m = \frac{\nu_t}{K_m g}$$

with :

- L : Structural volumetric length (cm)
- $[\dot{p}_{M,t}]$: Volume-specific maintenance rate ($J.d^{-1}.cm^{-3}$)
- $[E_G]$: Volume-specific cost for structure ($J.cm^{-3}$)
- κ : Energy allocation fraction to soma (-)

$$W_W = L^3 \left(1 + \frac{E}{E_m} w \right)$$

with :

- W_W : Wet weight of the earthworm (g)
- w : Contribution of reserve to body weight ($g.cm^{-3}$)

Maturity and reproduction

$$\dot{p}_C = \frac{g}{g + \frac{E}{E_m}} (\nu_t L^2 + K_m L^3)$$

with :

- $\dot{p}_C$: Mobilized energy flux from reserve ($J.d^{-1}$)

$$\dot{p}_R = (1 - \kappa) \dot{p}_C - \dot{k}_{J,t} E_h$$

with :

- $\dot{p}_R$: Allocated energy flux to reproduction ($J.d^{-1}$)
- $\dot{k}_{J,t}$: Maturity maintenance rate (d^{-1})

$$\frac{dE_H}{dt} = \begin{cases} E_H \leq E_H^p, & \dot{p}_R \\ E_H > E_H^p, & 0 \end{cases}$$

with :

- E_h : Maturity (J)
- E_h^p : Maturity at puberty (J)

$$\frac{dR}{dt} = \begin{cases} E_H \leq E_H^p, & 0 \\ E_H > E_H^p, & \kappa_R \frac{\dot{p}_R}{E_0} \end{cases}$$

with :

- R : Cumulated reproduction (#)
- κ_R : Reproduction efficiency (-)
- E_0 : Energy cost for an egg (J)

Effect of temperature on rates

$$s_A = \exp\left(\frac{T_A}{T_{ref} - \frac{T_A}{T_{exp} + 273.15}}\right)$$

$$s_{rH} = \frac{1 + \exp\left(\frac{T_A^H}{T^H} - \frac{T_A^H}{T_{ref} + 273.15}\right)}{1 + \exp\left(\frac{T_A^H}{T^H} - \frac{T_A^H}{T_{exp} + 273.15}\right)}$$

$$T_c = \begin{cases} \text{if } T_{exp} + 273.15 \geq T_{ref}, & s_A \\ \text{if } T_{exp} + 273.15 < T_{ref}, & s_{rH} \end{cases}$$

$$\{\dot{p}_{Am}\}_t = \{\dot{p}_{Am}\} T_c$$

$$[\dot{p}_{M,t}] = [\dot{p}_M] T_c$$

$$\nu_t = \nu T_c$$

$$\dot{k}_{J,t} = \dot{k}_J T_c$$

with :

- T_A : Arrhenius temperature (K)
- T_A^H : Arrhenius temperature for upper boundary (K)
- T^H : Upper boundary (K)
- T_{ref} : Reference temperature (K)
- T_{exp} : Experimental temperature (°C)

2.2 Priors used

Priors are the same between models :

- Distrib (Fm, LogUniform, 0.0001, 1);
- Distrib (pAm, TruncNormal_cv, 100, 0.2, 40, 2000);
- Distrib (r_pAm_pM, TruncNormal_cv, 0.8, 0.1, 0.1, 2);
- Distrib (v, TruncNormal_cv, 0.18505, 0.1, 0.001, 2);
- Distrib (kap, Uniform, 0, 1);
- Distrib (kapJ, Uniform, 0, 1);
- Distrib (kapA, Uniform, 0, 1);
- Distrib (Ehp, TruncNormal_cv, 100, 0.2, 20, 300);
- Distrib (E_coc, TruncNormal_cv, 30, 0.2, 1, 300);
- Distrib (rOM_ClxHorse, LogUniform, 0.00001, 0.3);
- Distrib (K, Uniform, 0.0001, 1);
- Distrib (dv, TruncNormal_cv, 1, 0.05, 0, 4);
- Distrib (wE, LogUniform, 1, 1000);

- Distrib (de, LogUniform, 0.000000001, 0.1);
- Distrib (tau, TruncNormal_cv, 7, 0.5, 0, 50);

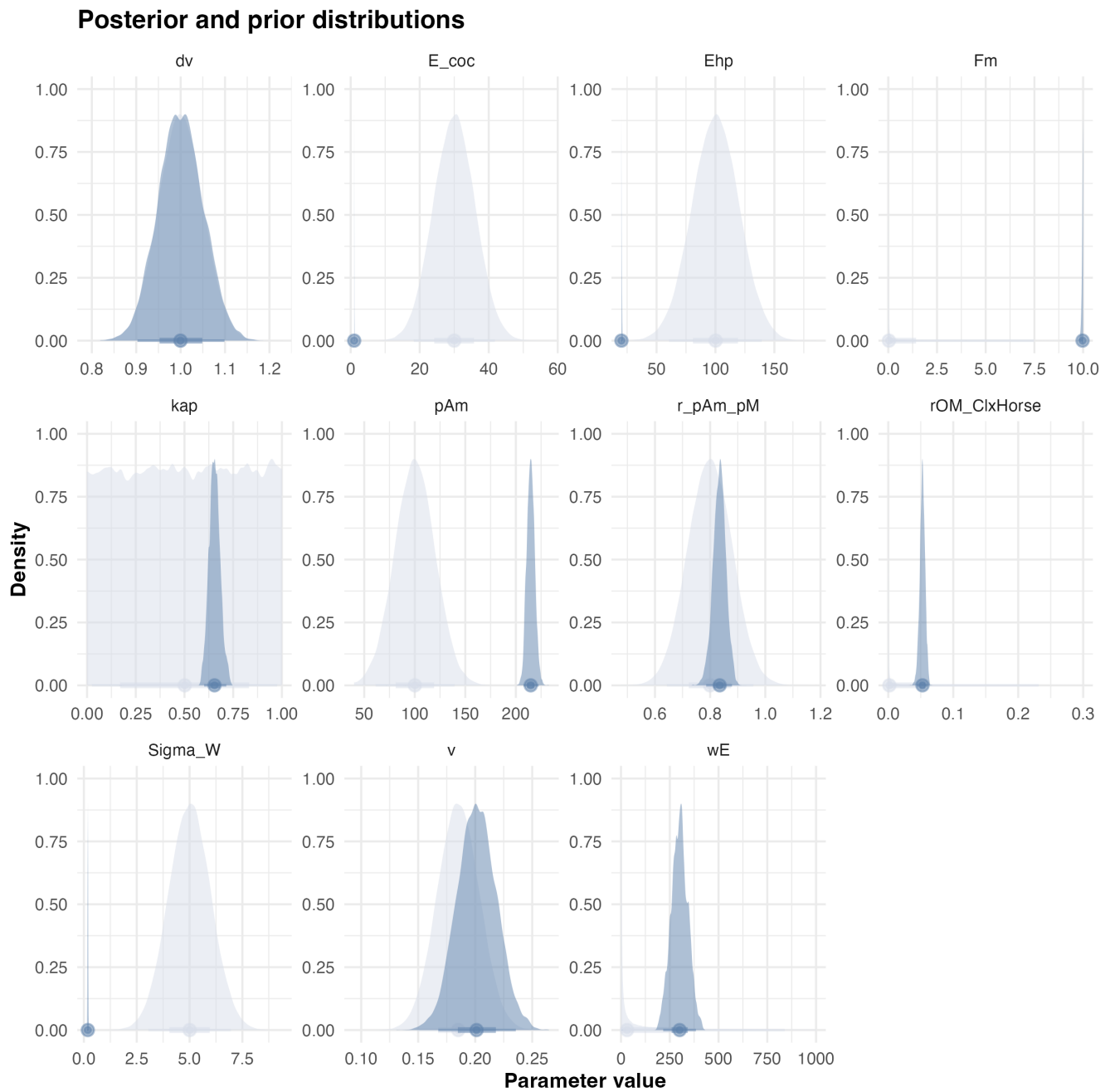
Log-Likelihood calculations :

- Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);
- Likelihood (Weight, Normal, Prediction(Weight), Sigma_W);
- Likelihood (Reproduction, Poisson, Prediction(Reproduction));

For the maximum ingestion rate, we imposed a biologically realistic constraint based on the assumption that an earthworm cannot ingest more than its own body weight per day. This constraint was used to bound the parameter $[\dot{F}_m]$.

An estimate of cocoon energy content was required to parameterize reproduction in the DEB model for *Aporrectodea caliginosa*. As no direct measurements are available for this species, we relied on published calorimetric data for *Lumbricidae* reported by Cumminns & Wuycheck (1971), who measured the energy density of whole individuals across developmental stages. Reported mean energy densities were 5,690 cal.g⁻¹ dry weight ($n = 3$), corresponding to 19,098 J.g⁻¹, and 782 cal.g⁻¹ wet weight ($n = 1$), corresponding to 3,268 J.g⁻¹. Although these measurements do not specifically target cocoons, they provide a reasonable order-of-magnitude estimate for annelid biomass. Assuming a mean cocoon fresh mass of 10 mg, this yields an approximate energy content of 33 J per cocoon. Comparable values have been reported for soil invertebrates in other calorimetric studies (e.g., Driver et al. (1974); Eggleton & Schramm (2004)), supporting the use of this approximation as a prior for egg energy content.

2.3 Model A_FLYw



{fig-scap="Estimated posterior distributions for each parameter."}

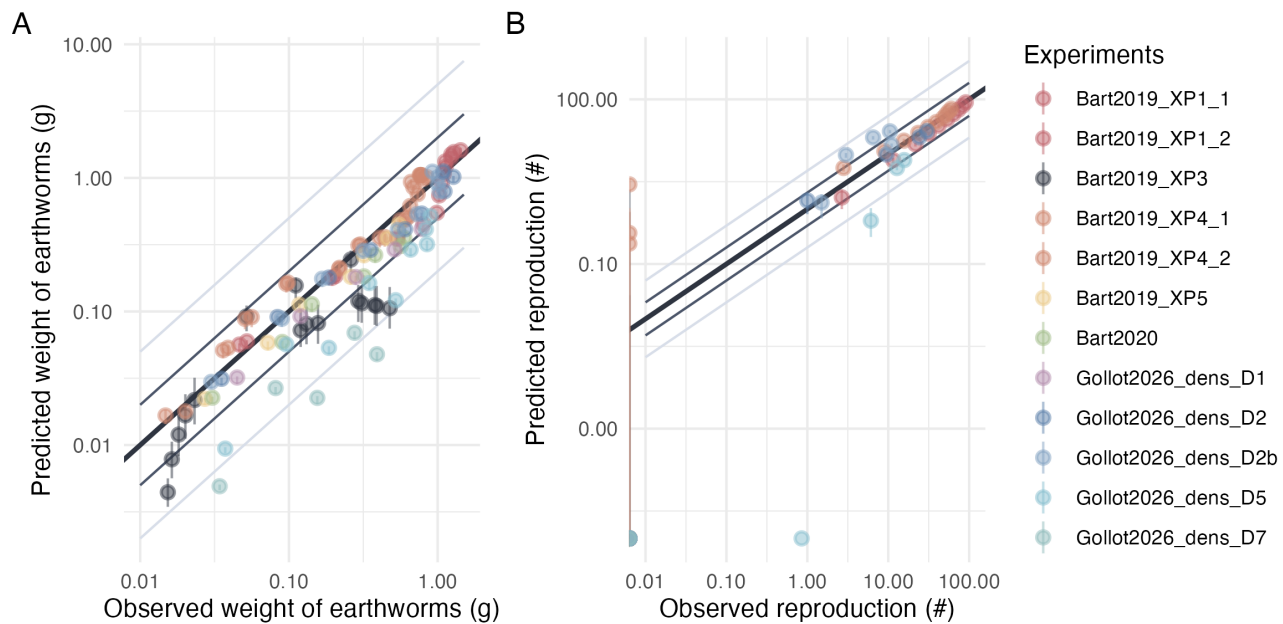


Figure 2.1: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

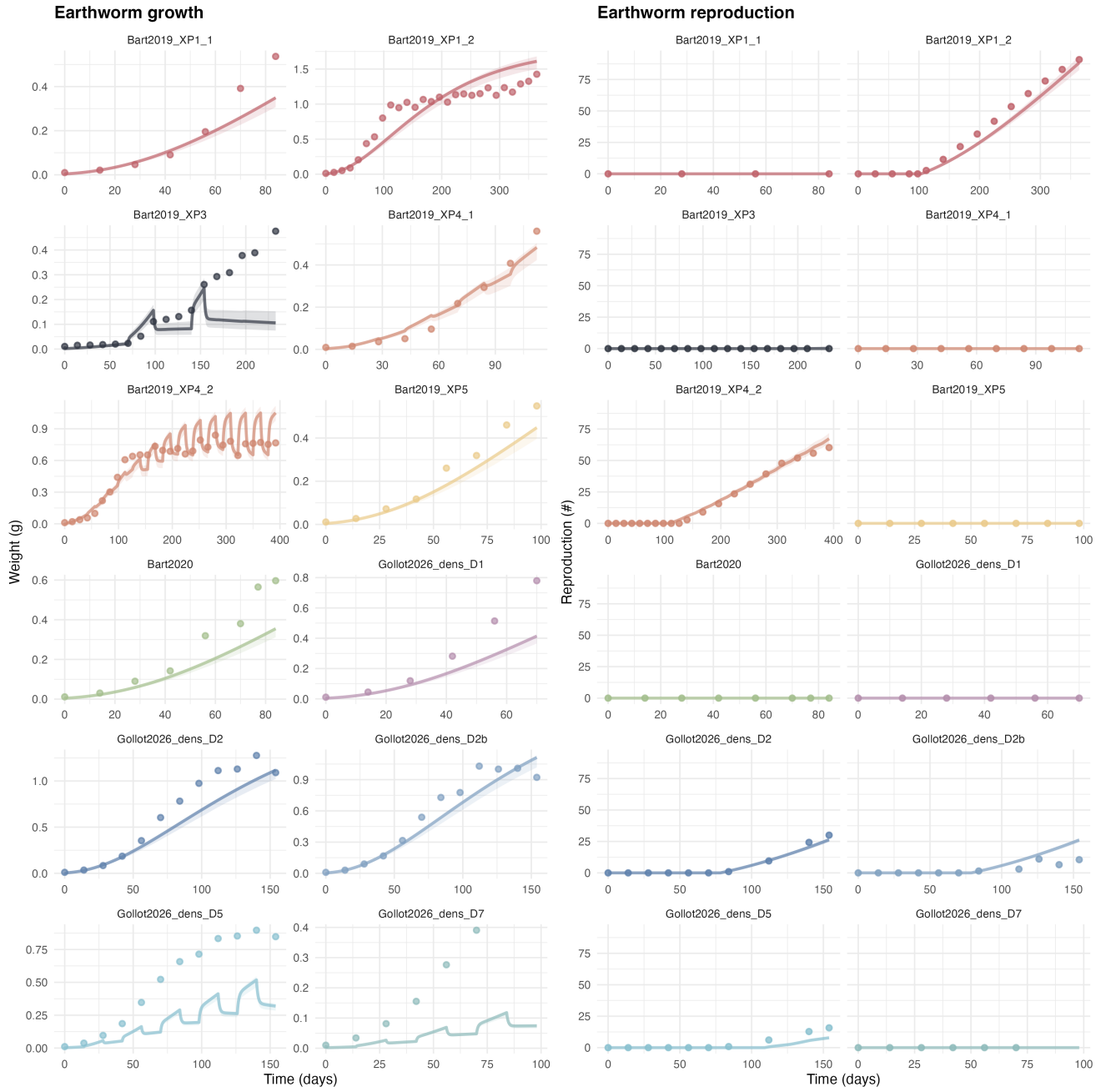


Figure 2.2: Observed and predicted growth and reproduction trajectories for each experiment. Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

2.4 Model B_FLYe

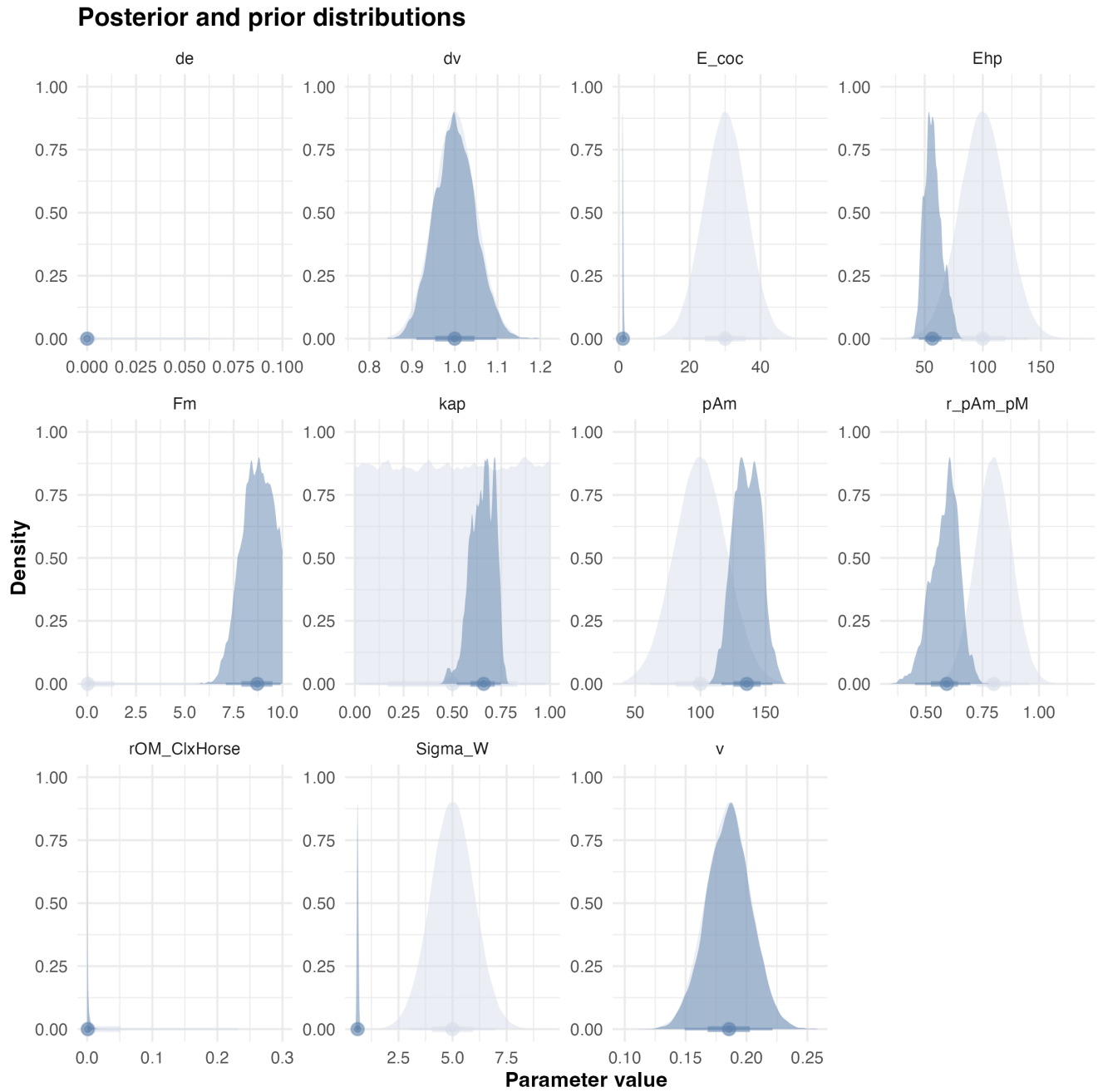


Figure 2.3: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

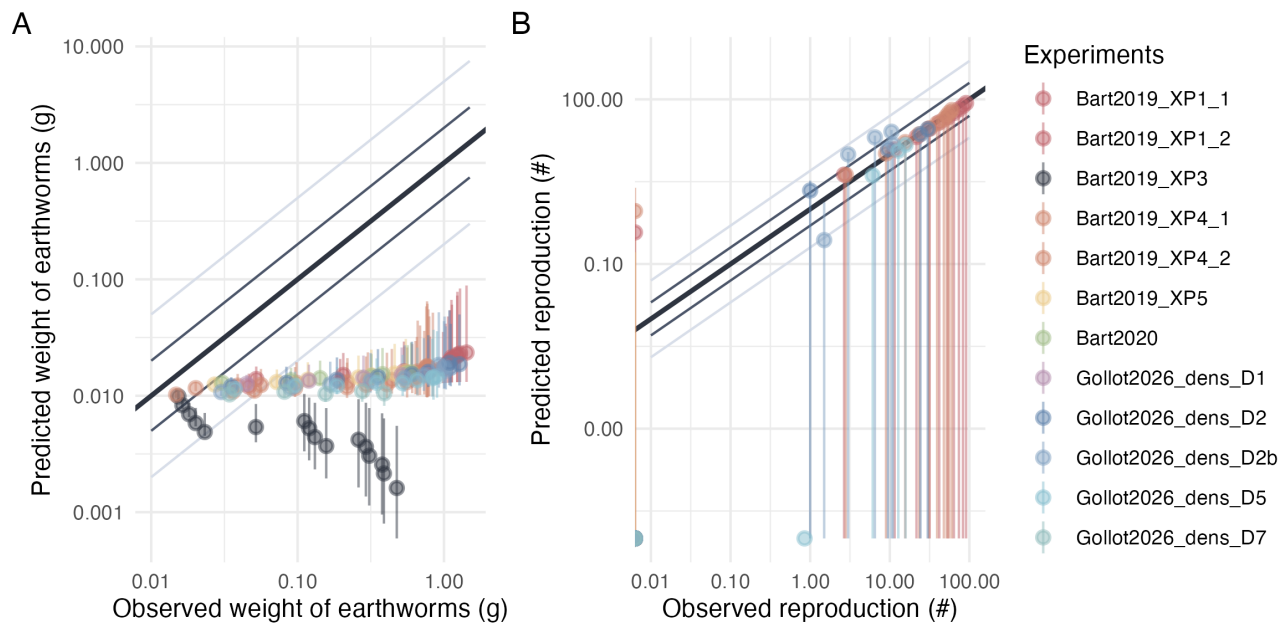


Figure 2.4: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

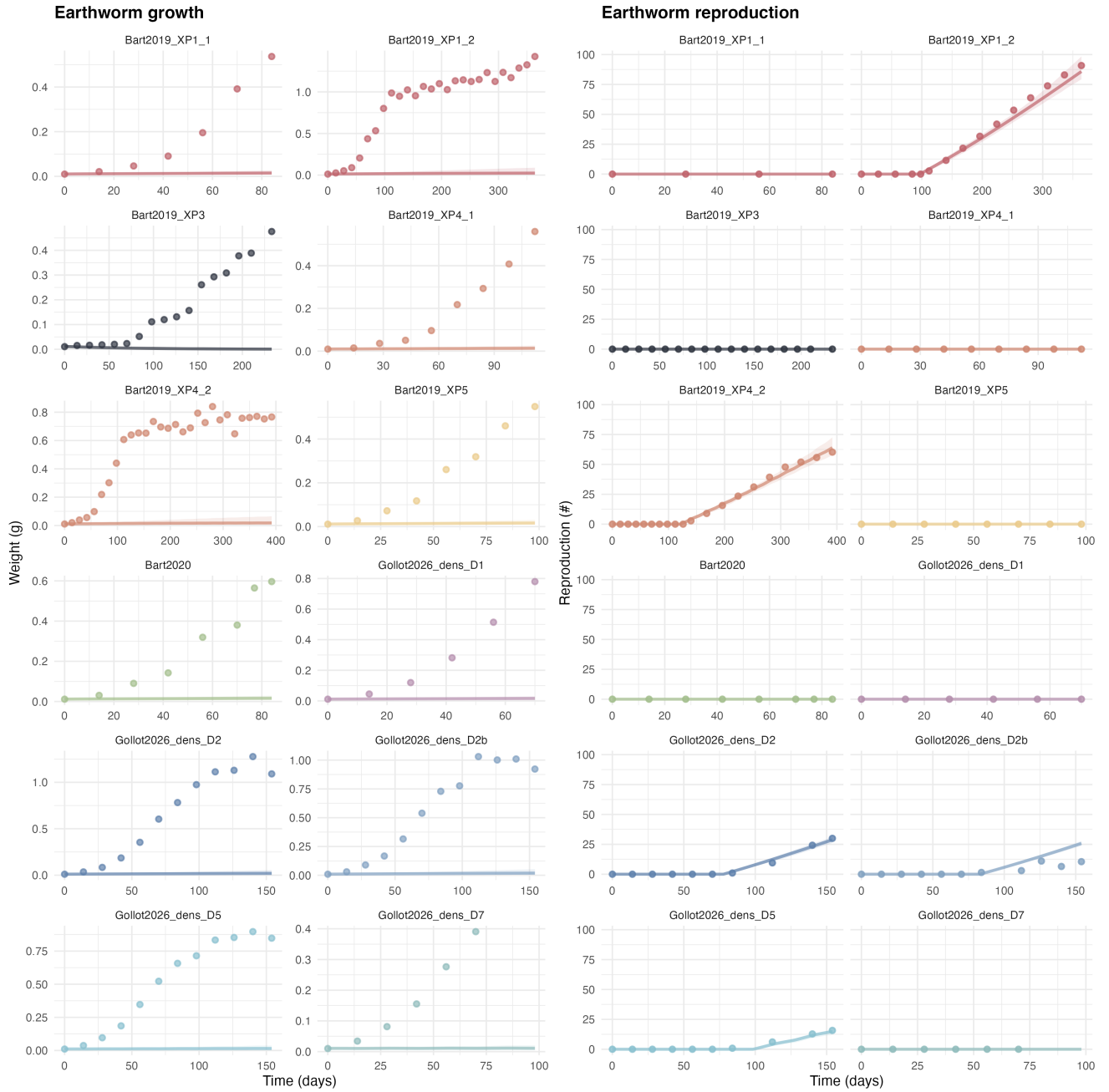


Figure 2.5: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

2.5 Model C_CWYe

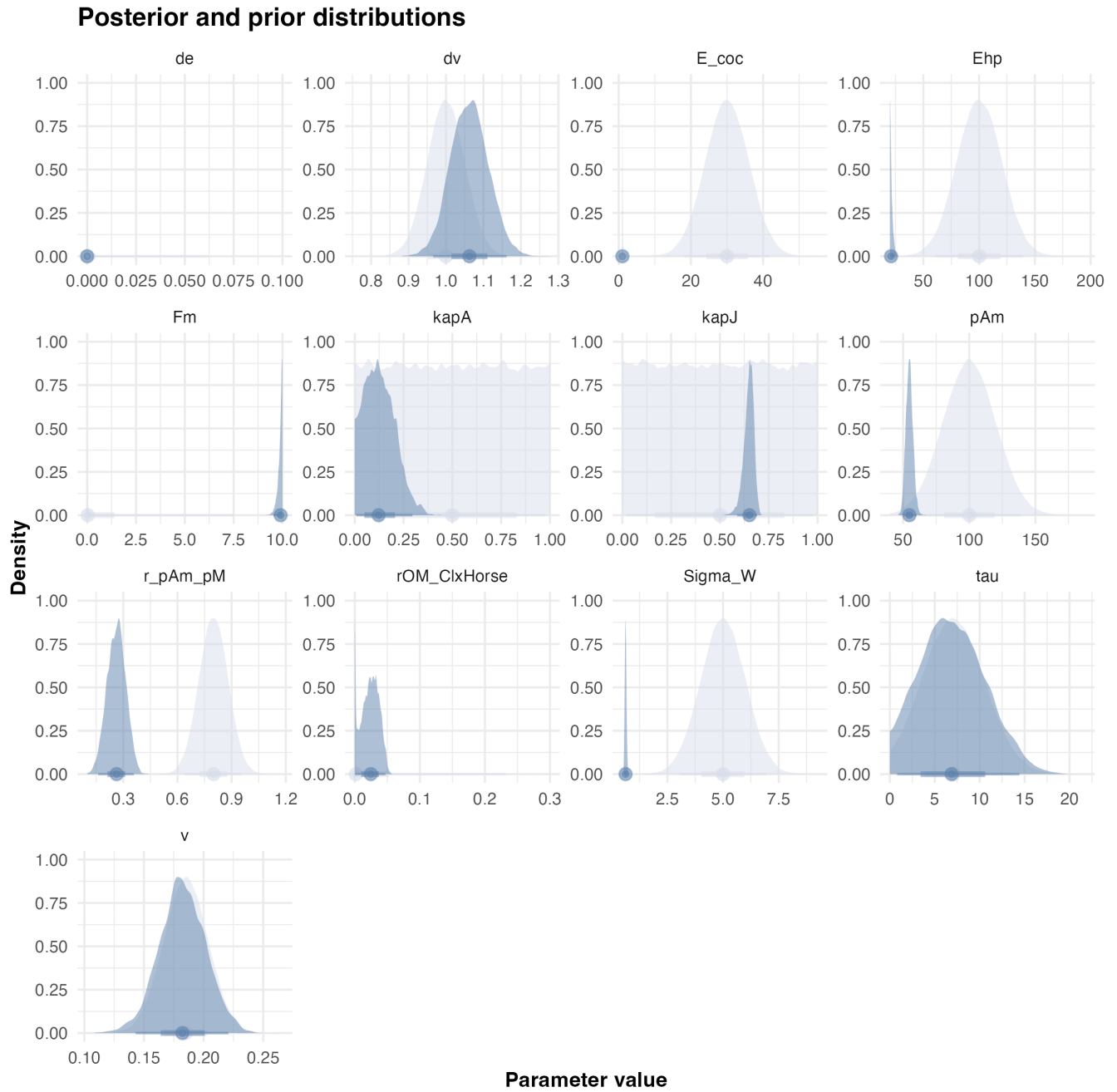


Figure 2.6: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

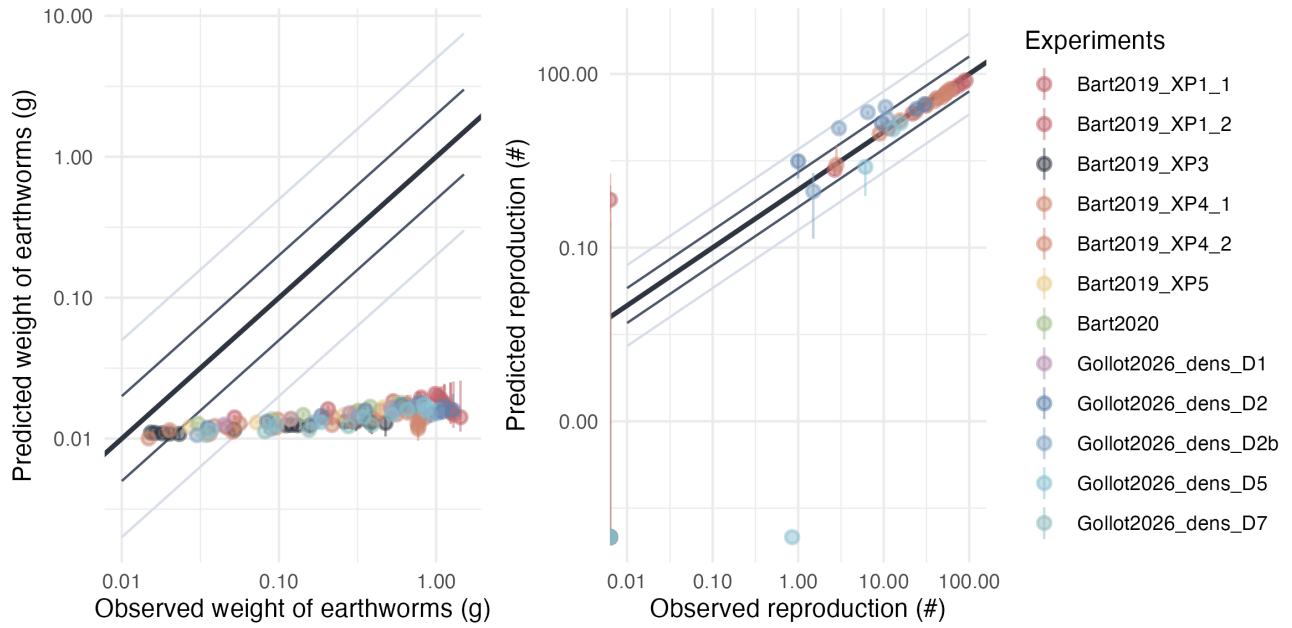


Figure 2.7: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

2.6 Model D_CWKe

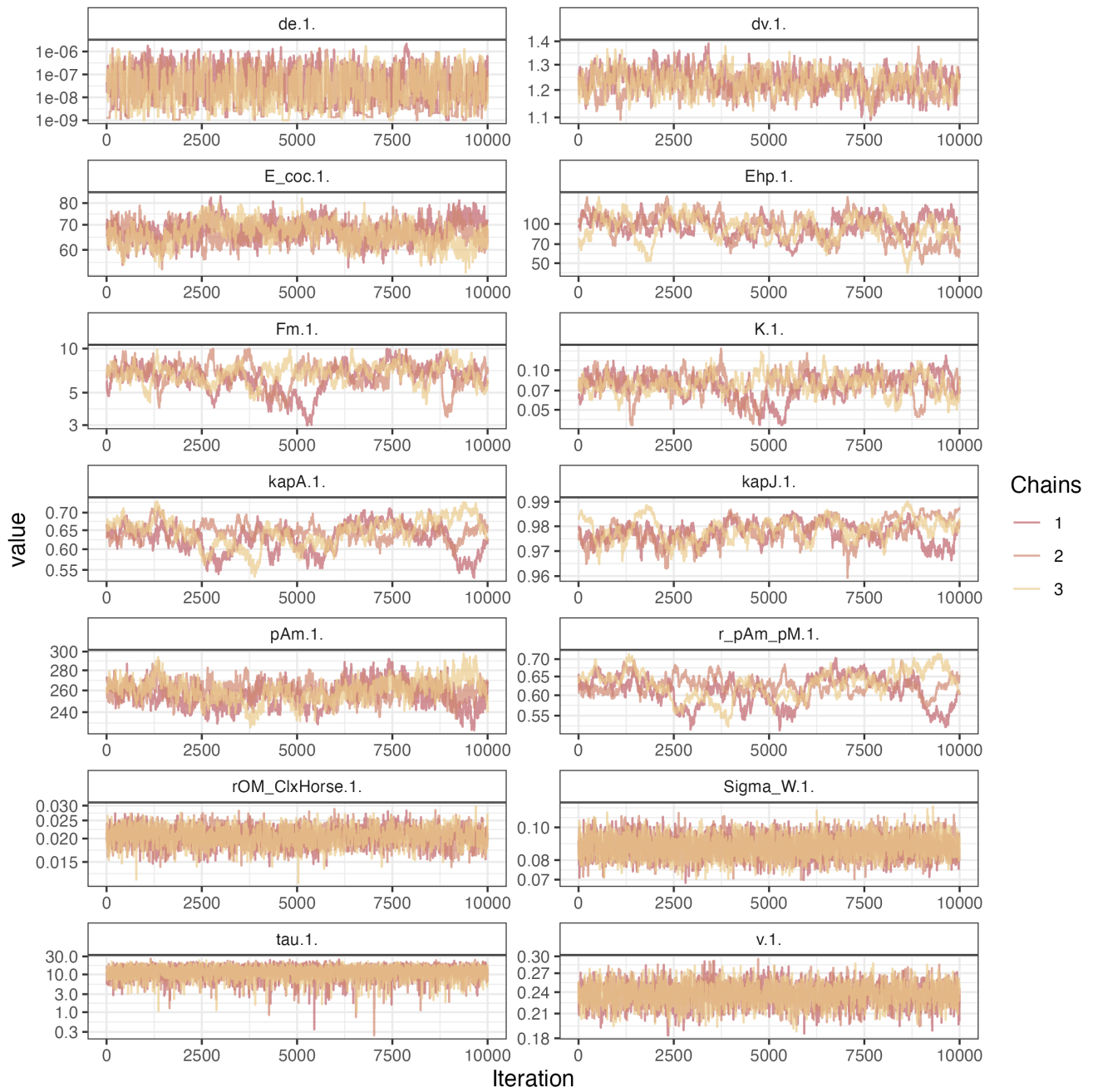


Figure 2.8: Model chains.

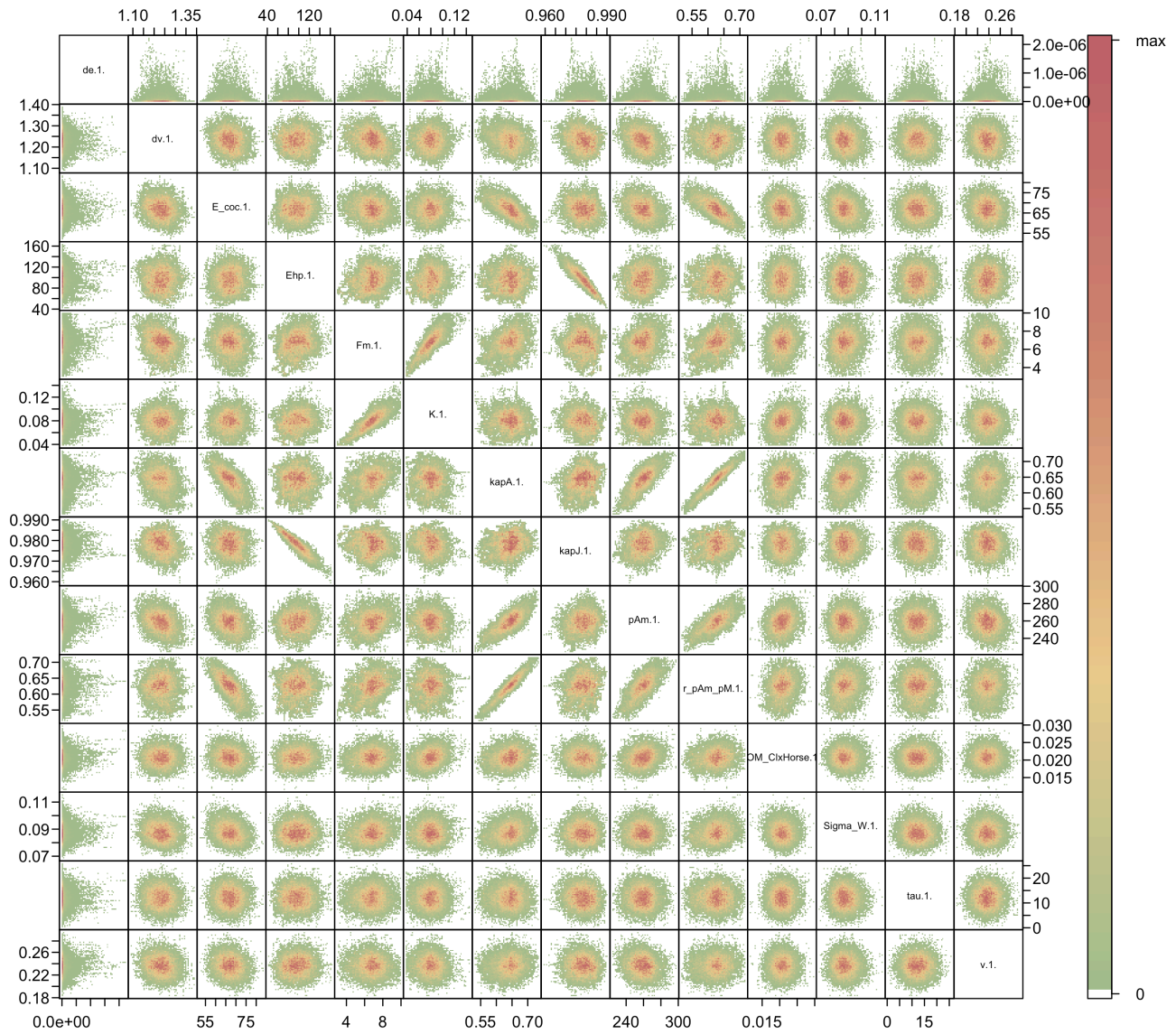


Figure 2.9: Chain correlations.

Posterior and prior distributions

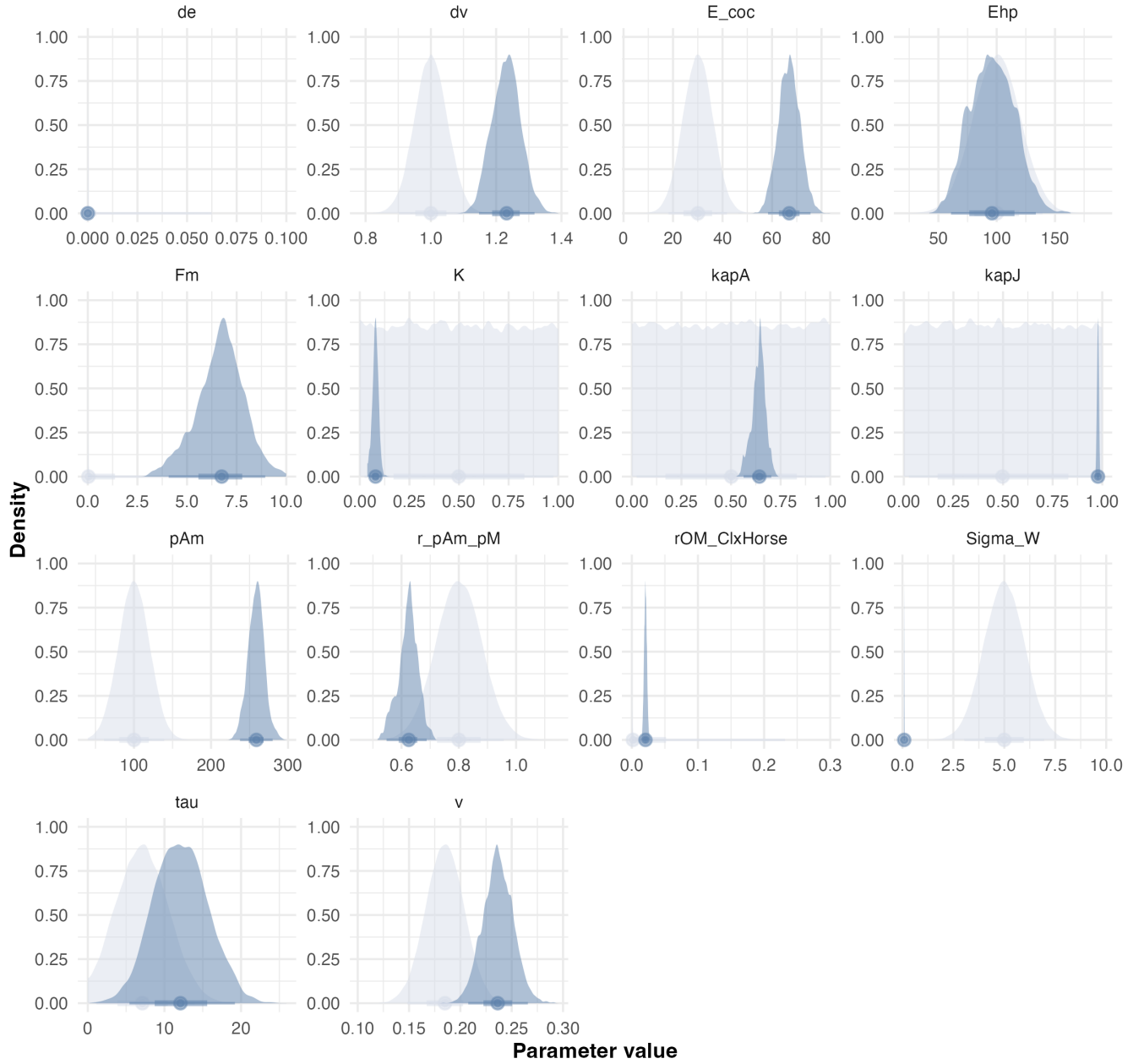


Figure 2.10: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

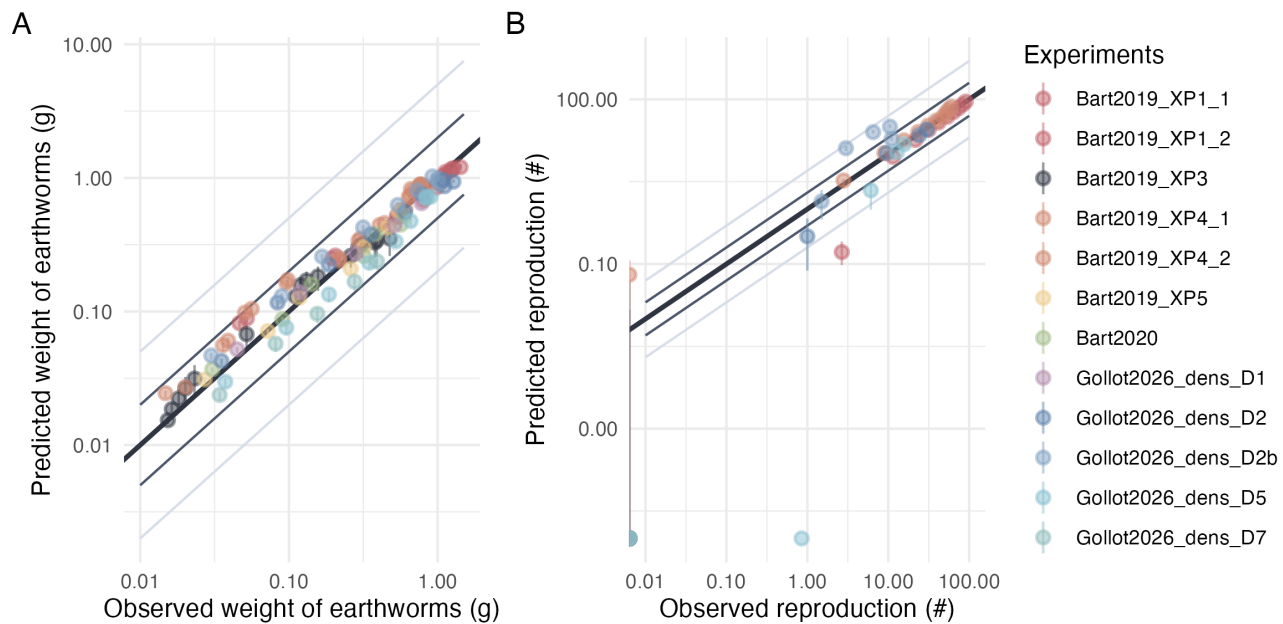


Figure 2.11: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

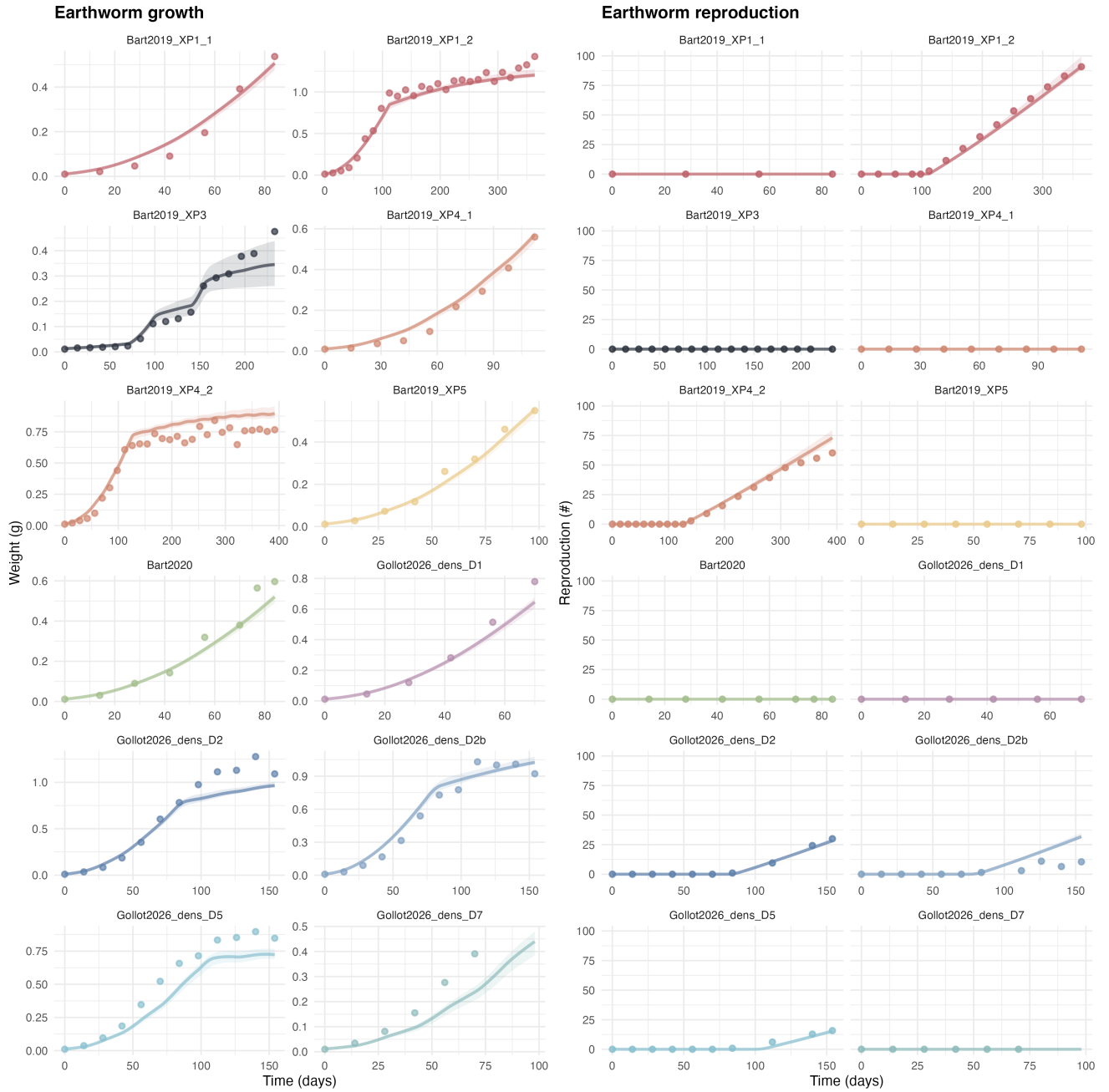


Figure 2.12: Observed and predicted growth and reproduction trajectories for each experiment. Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

2.7 Model E_CLYe

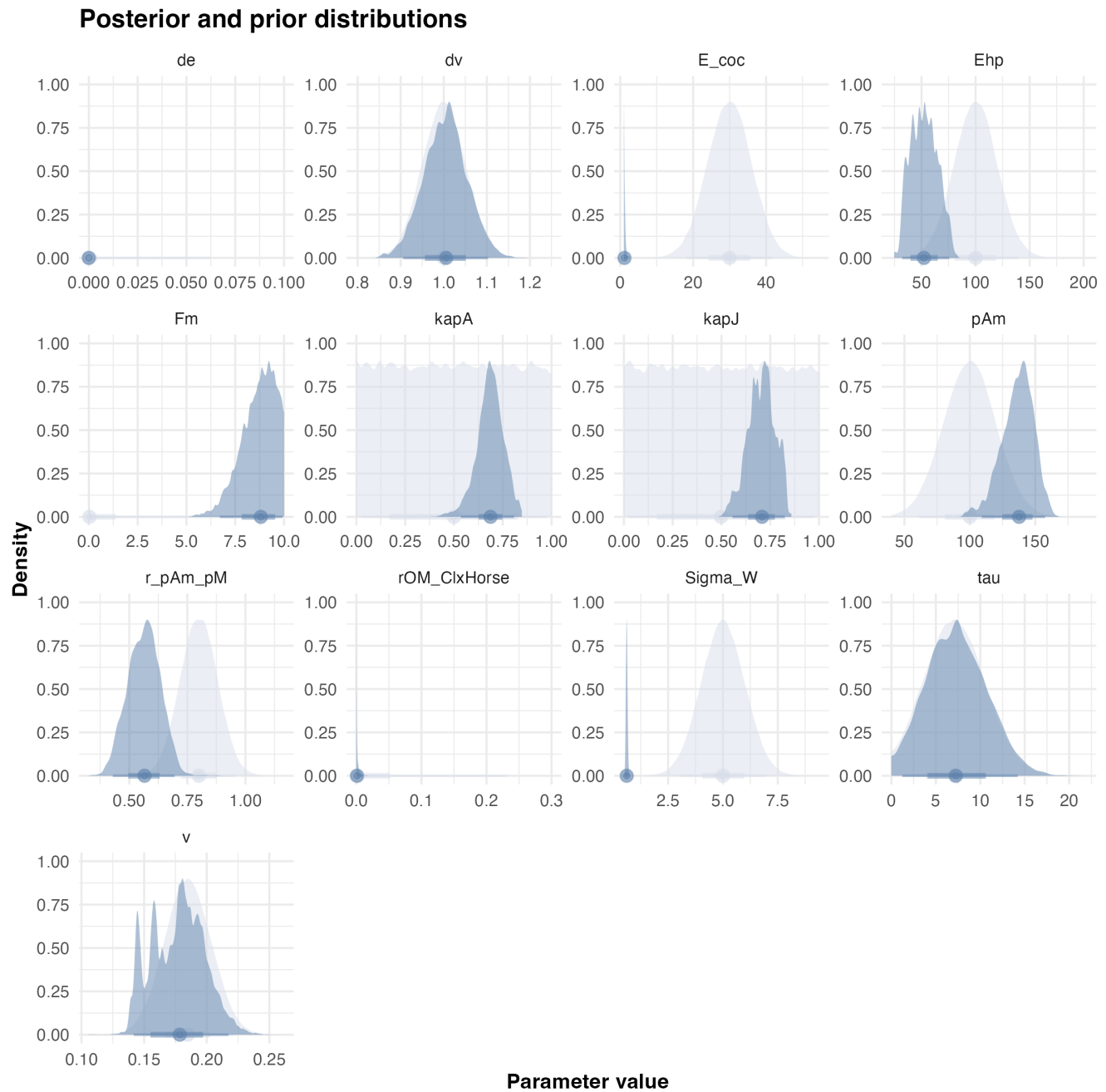


Figure 2.13: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

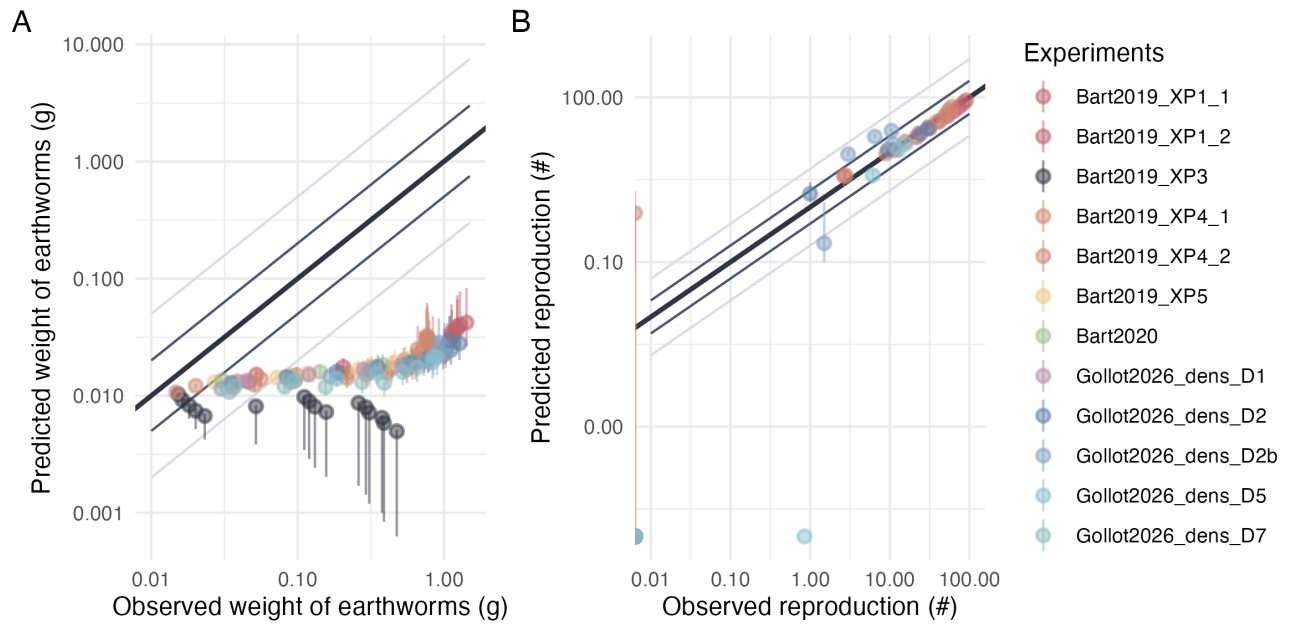


Figure 2.14: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

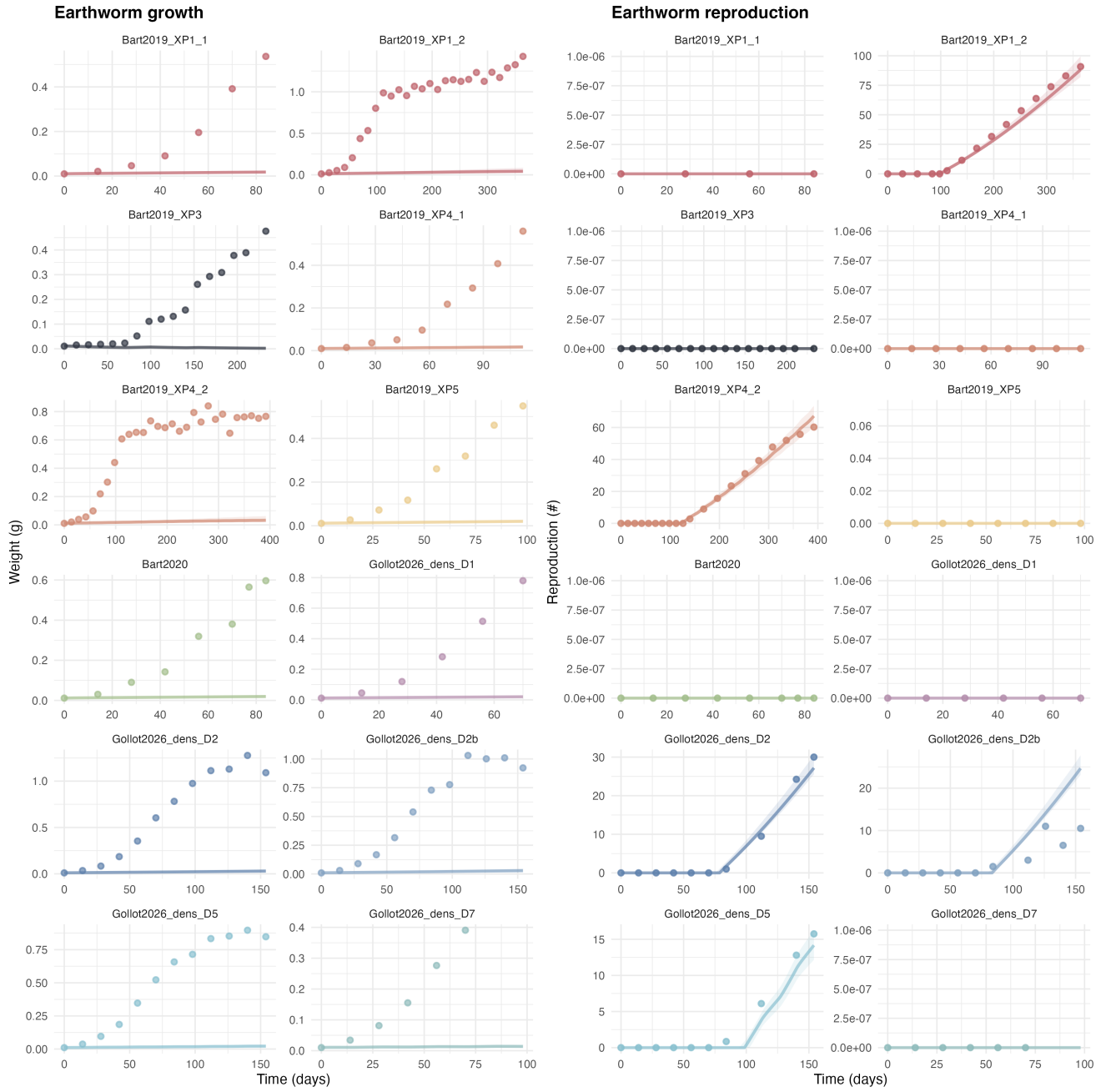


Figure 2.15: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

2.8 Model F_CLKe

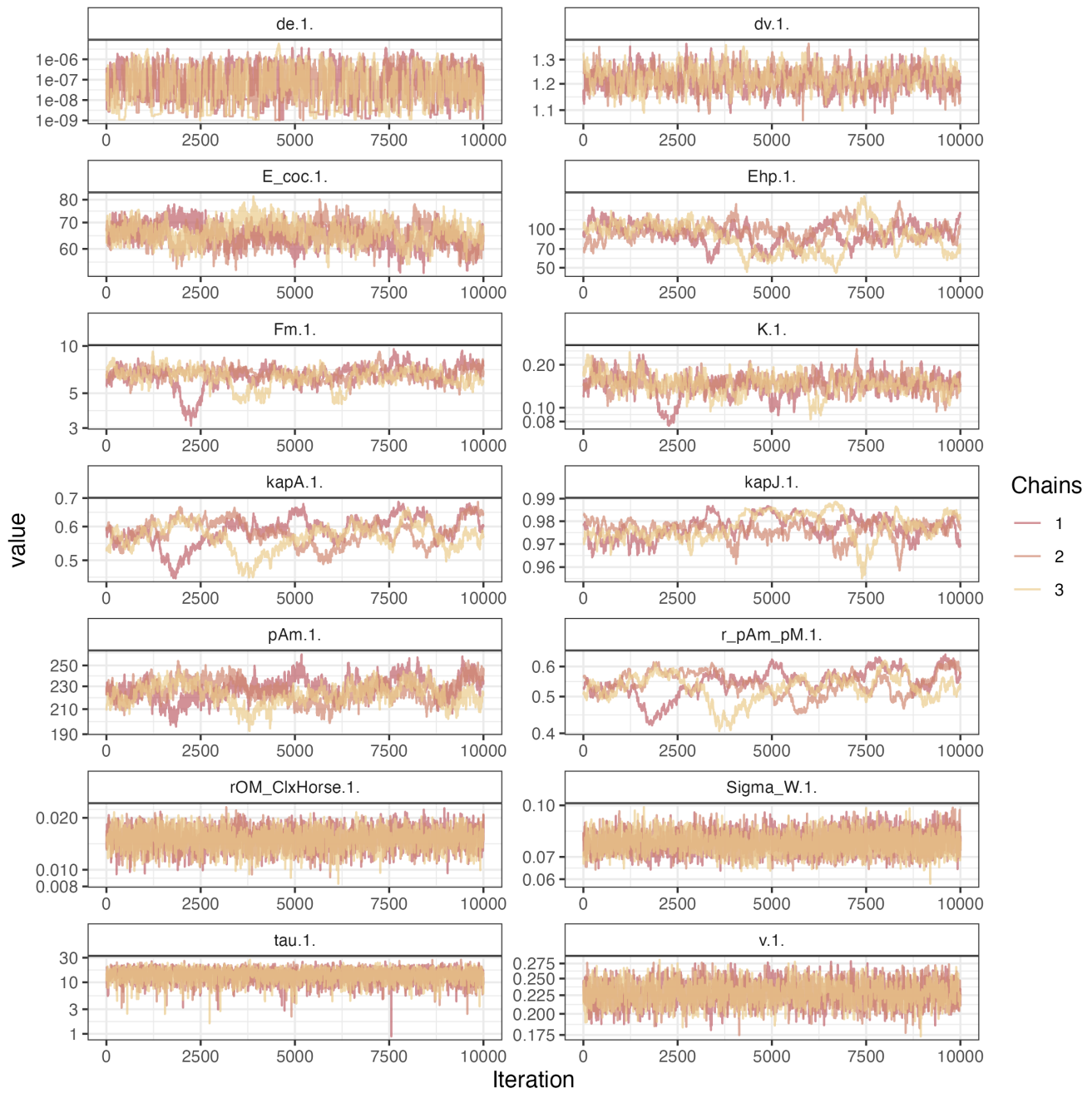


Figure 2.16: Model chains.

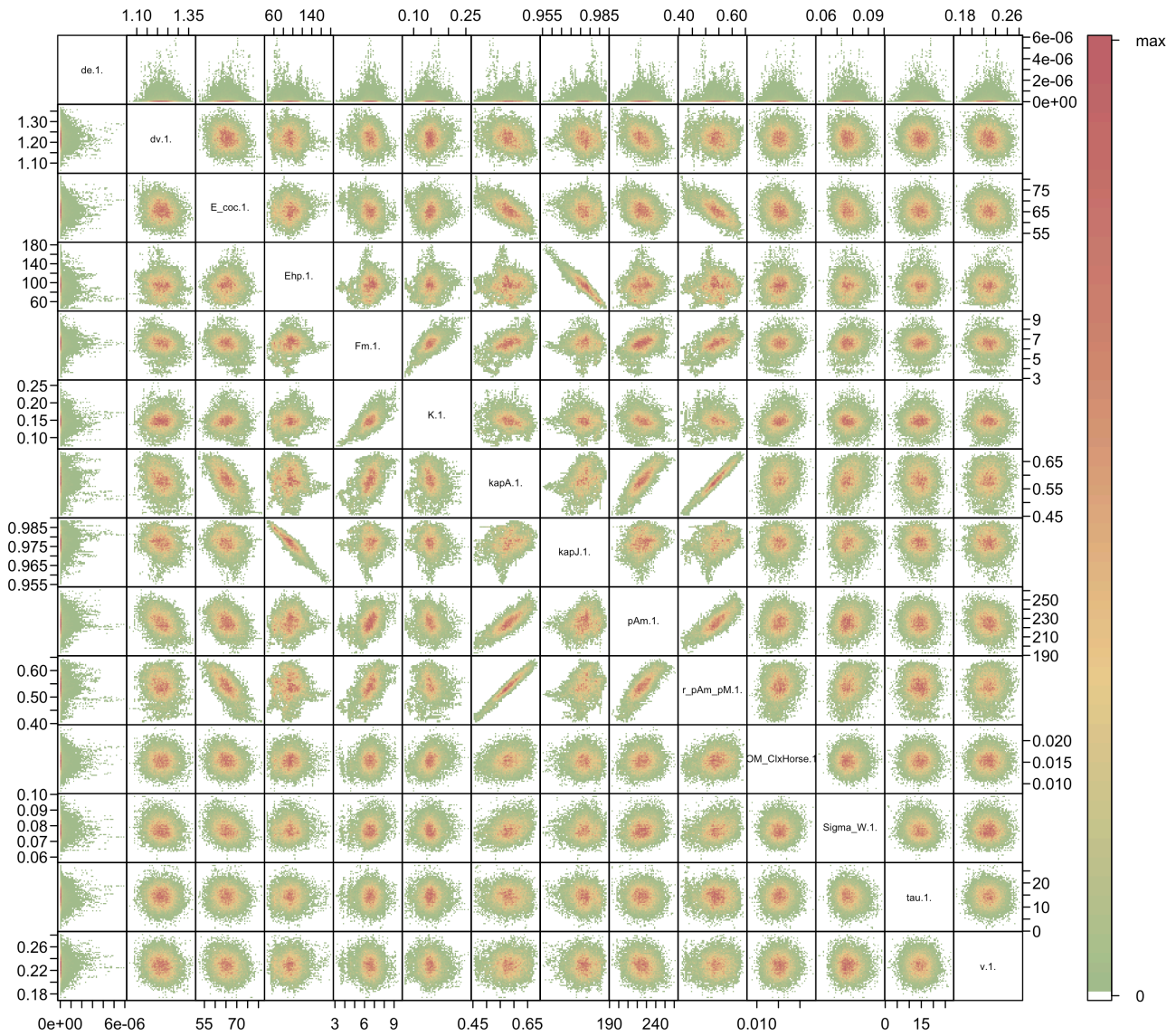


Figure 2.17: Chain correlations.

Posterior and prior distributions

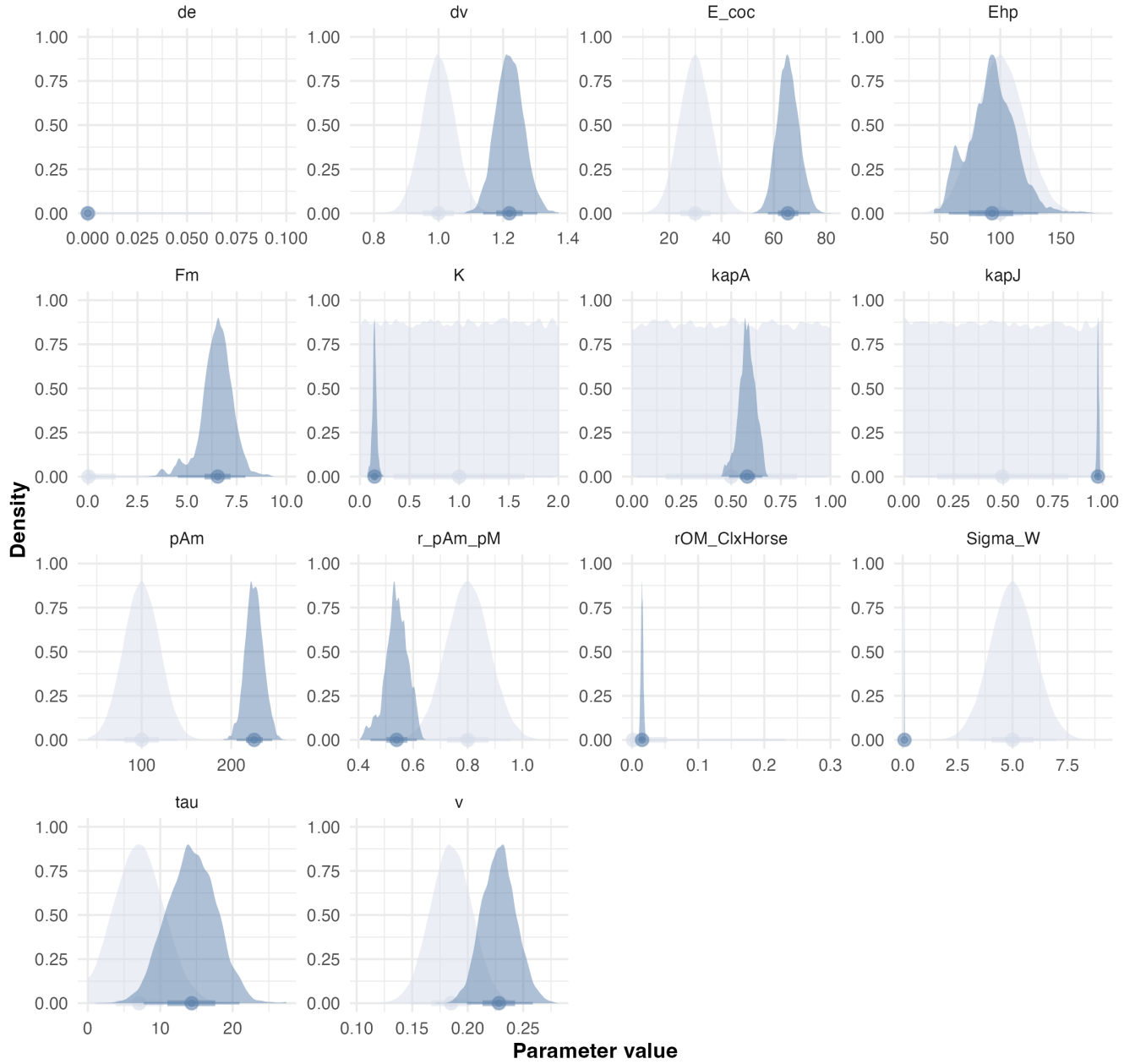


Figure 2.18: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

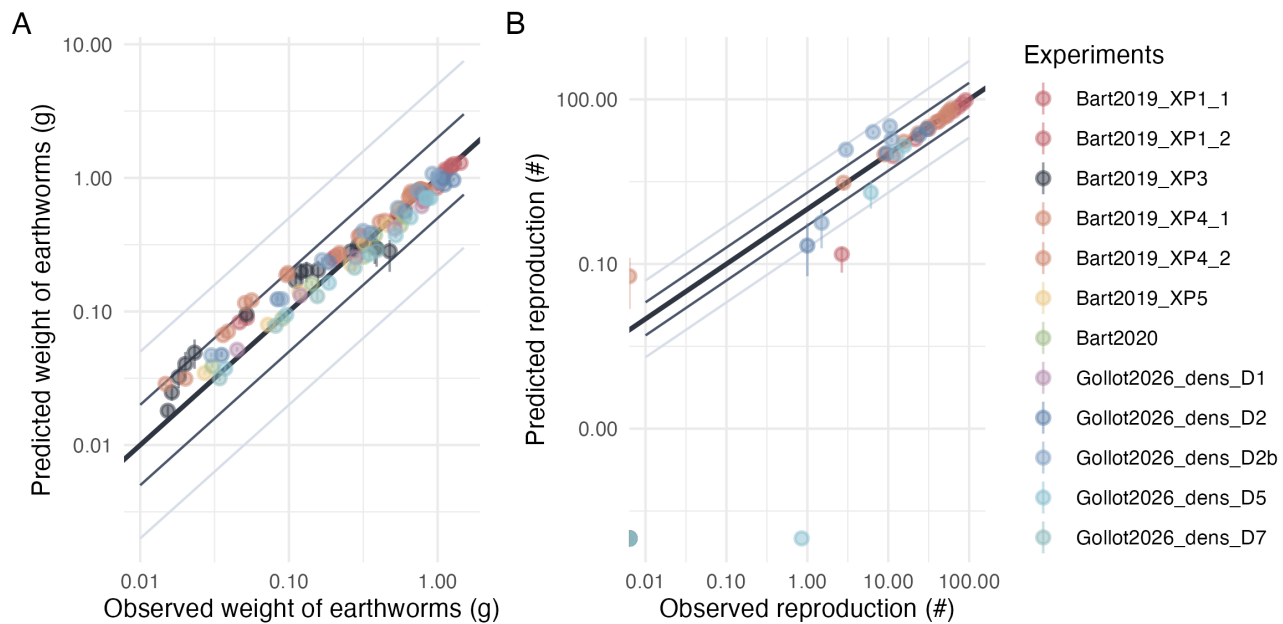


Figure 2.19: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

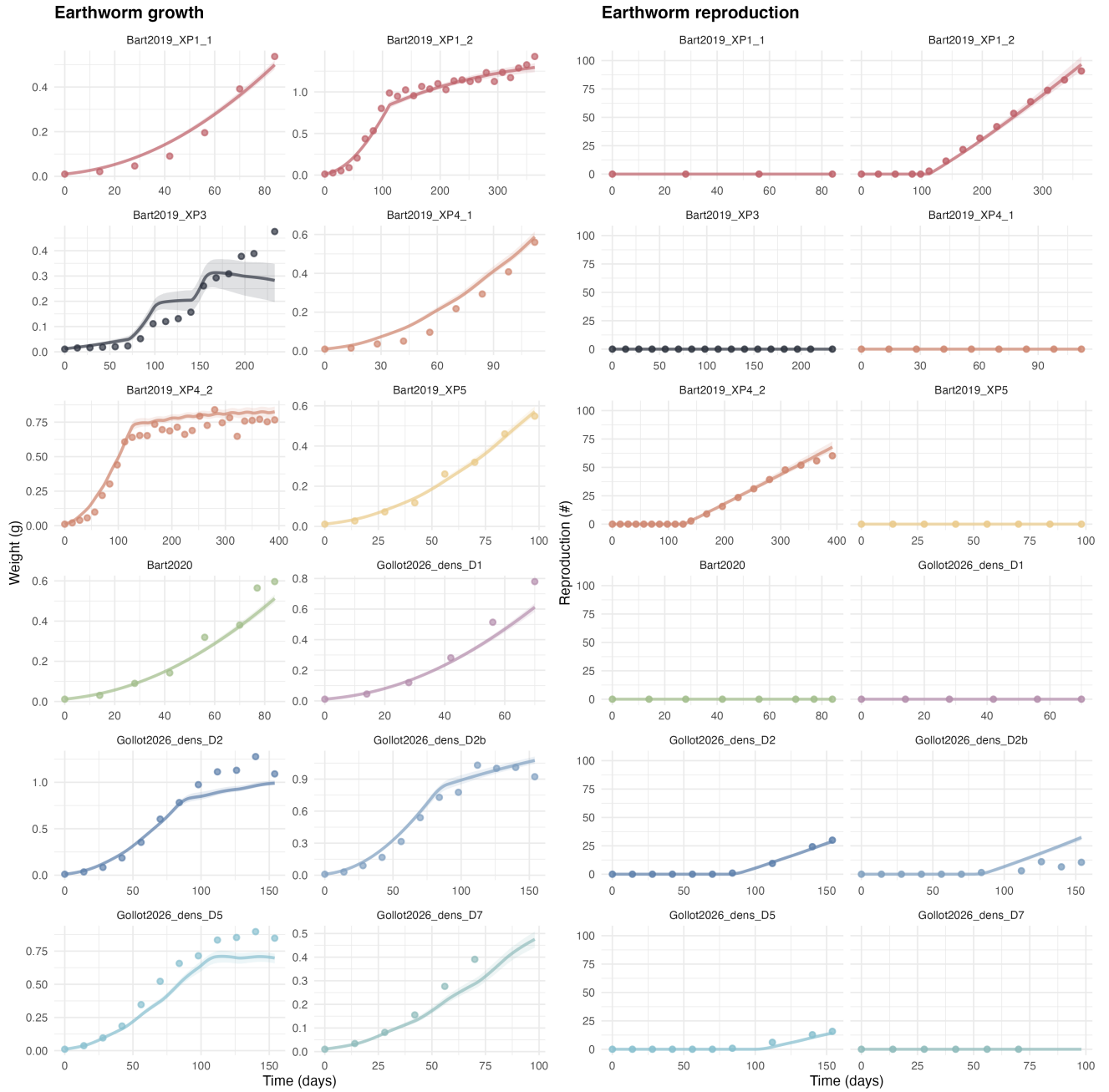


Figure 2.20: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

2.9 Model G_FLKe

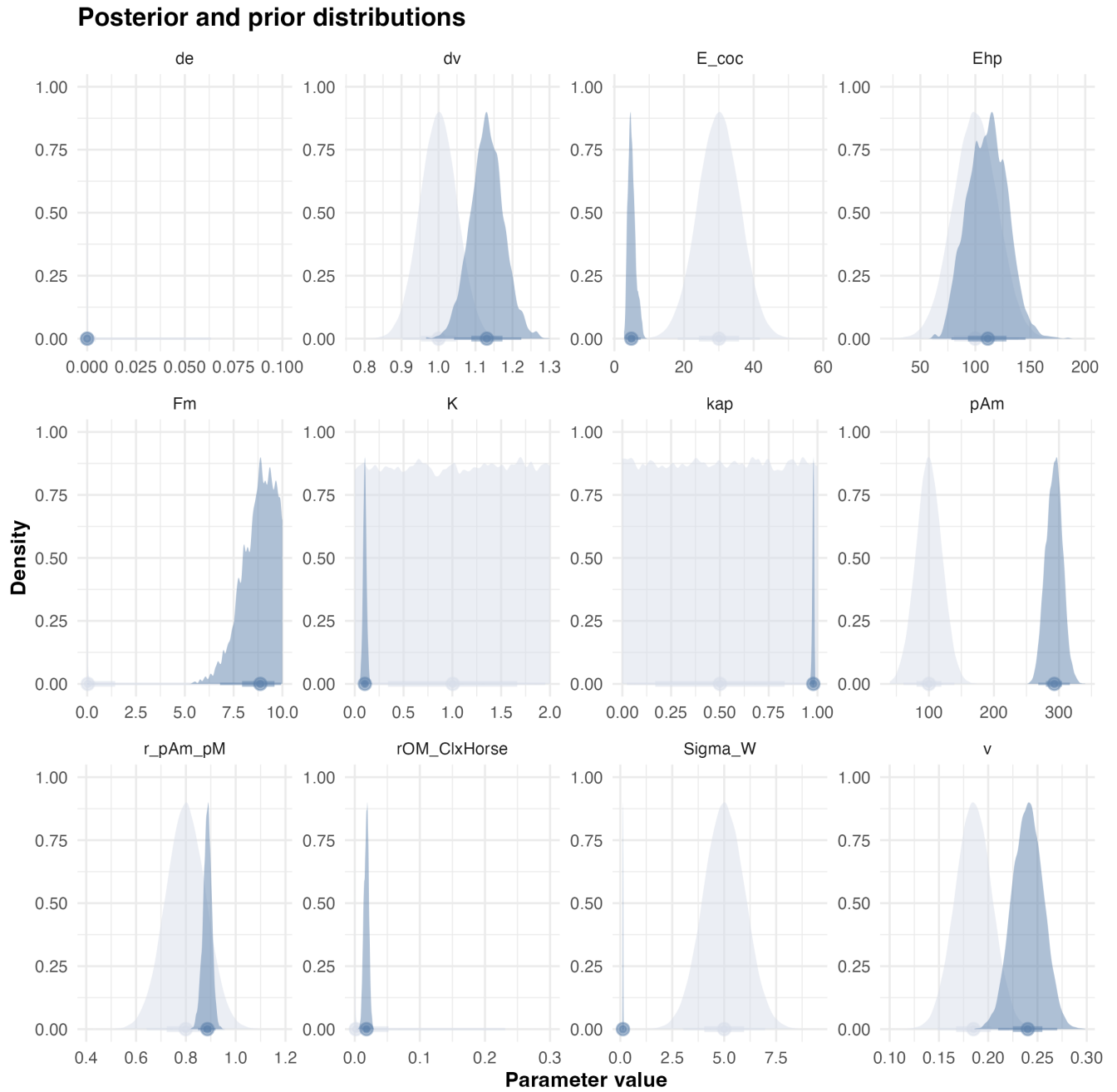


Figure 2.21: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

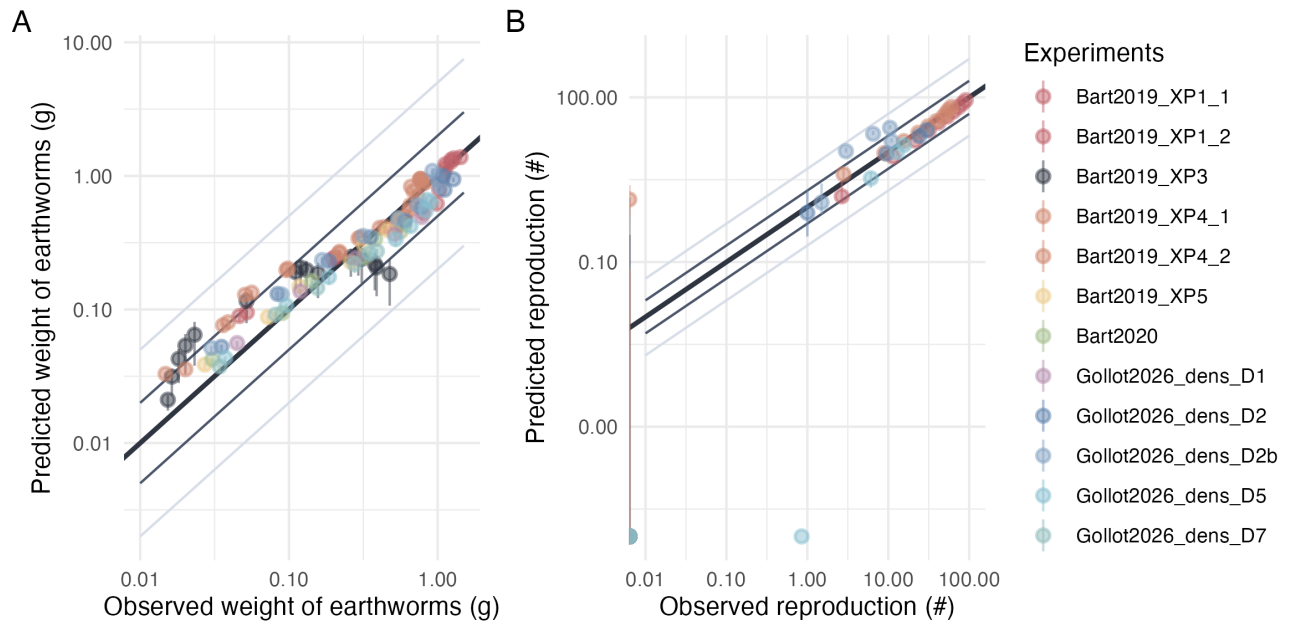


Figure 2.22: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

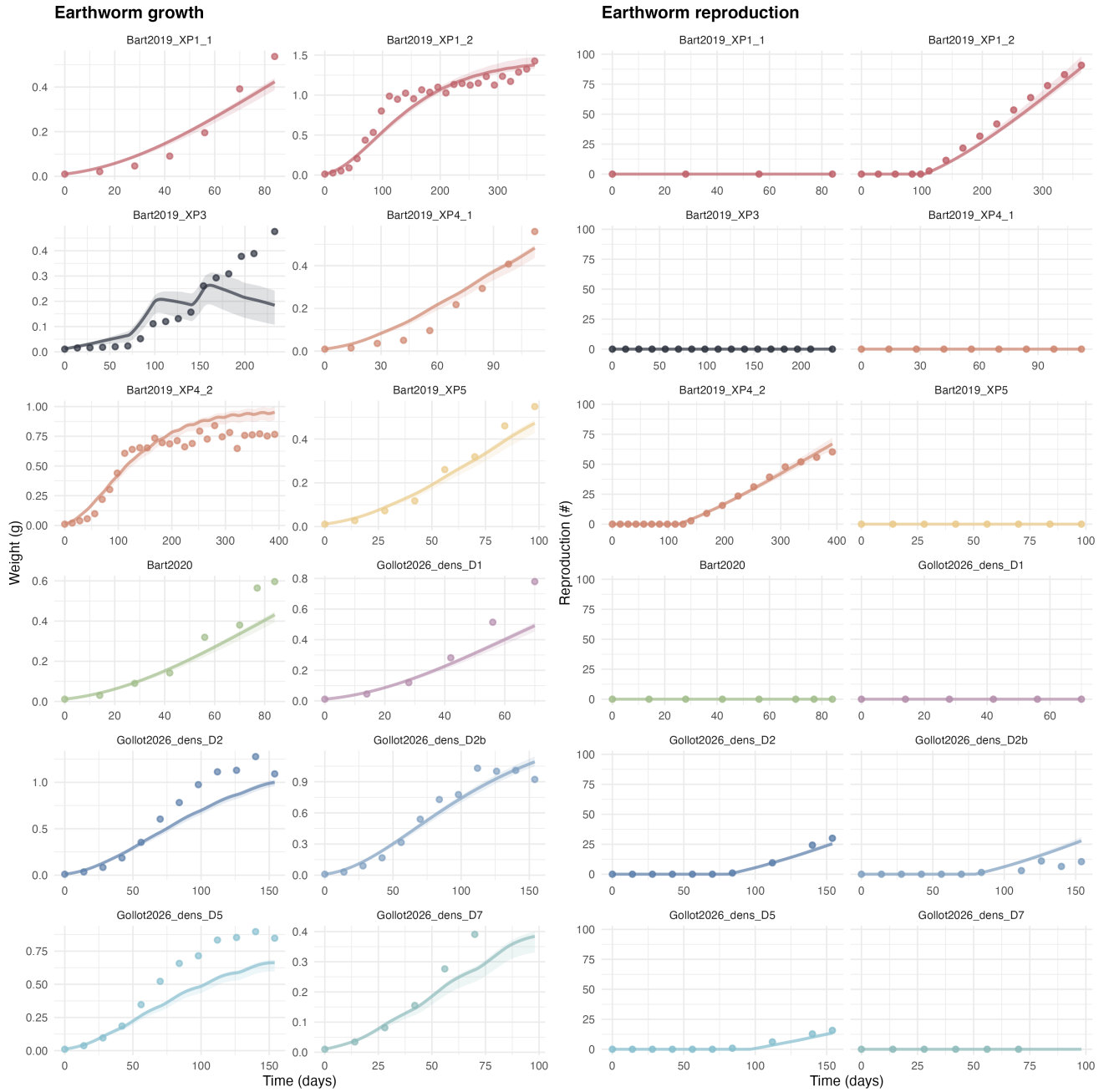


Figure 2.23: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

2.10 Model H_FWKe

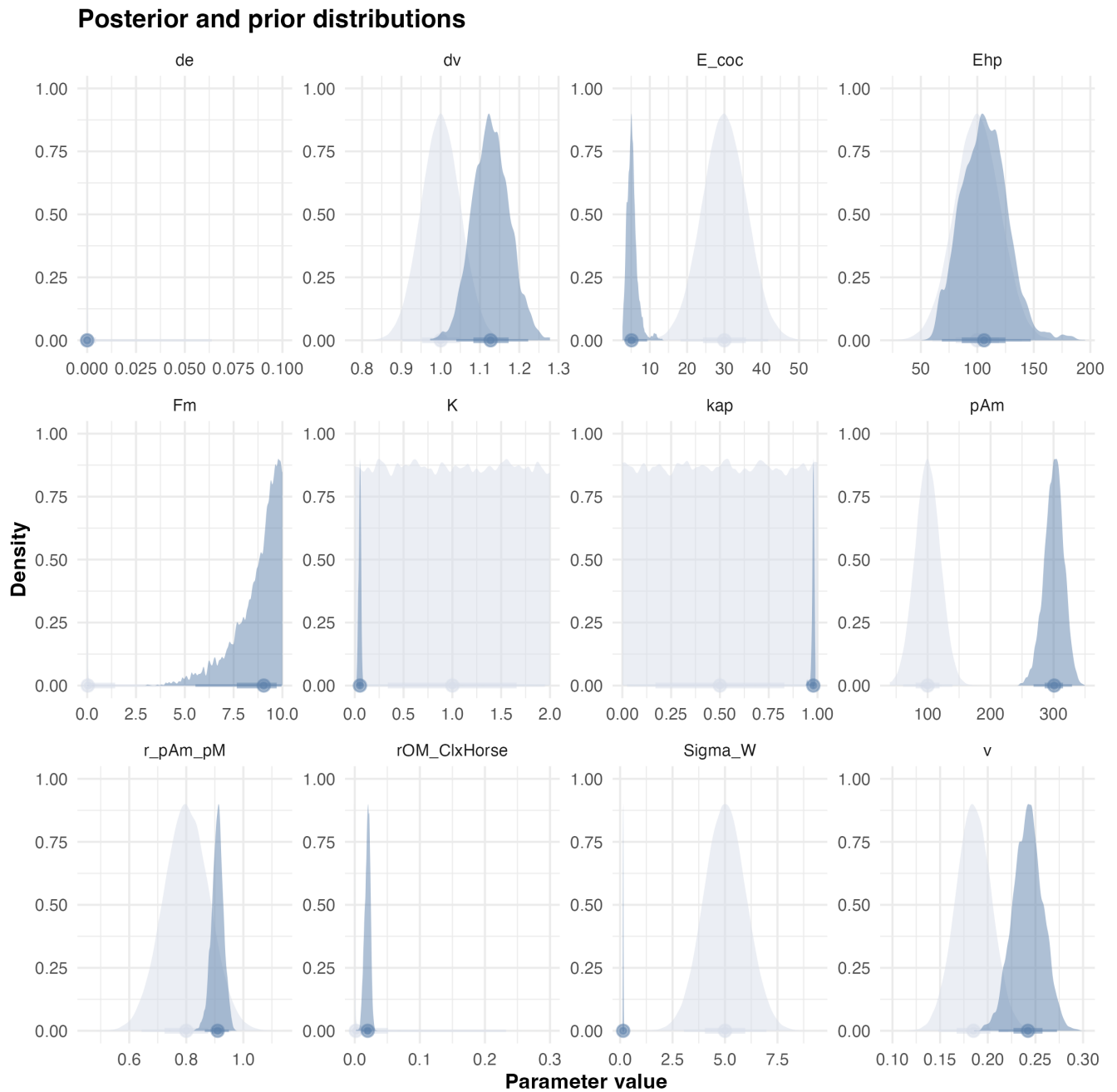


Figure 2.24: Estimated posterior distributions for each parameter. Prior and posterior distributions are represented in light and dark blue, respectively.

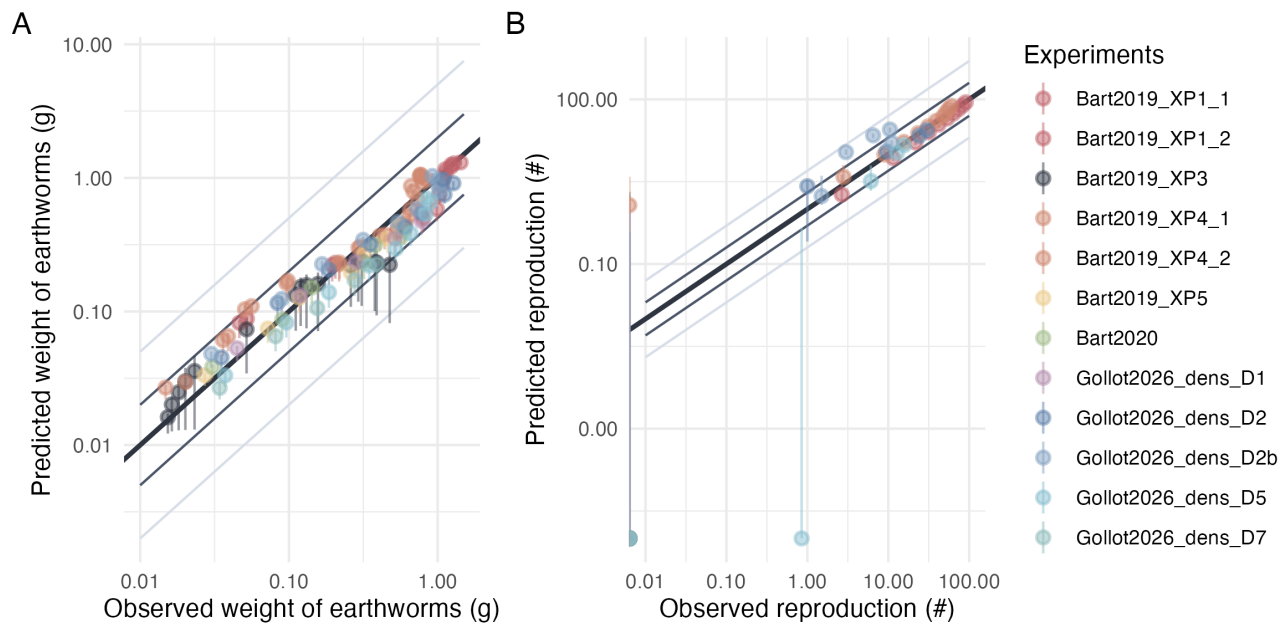


Figure 2.25: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

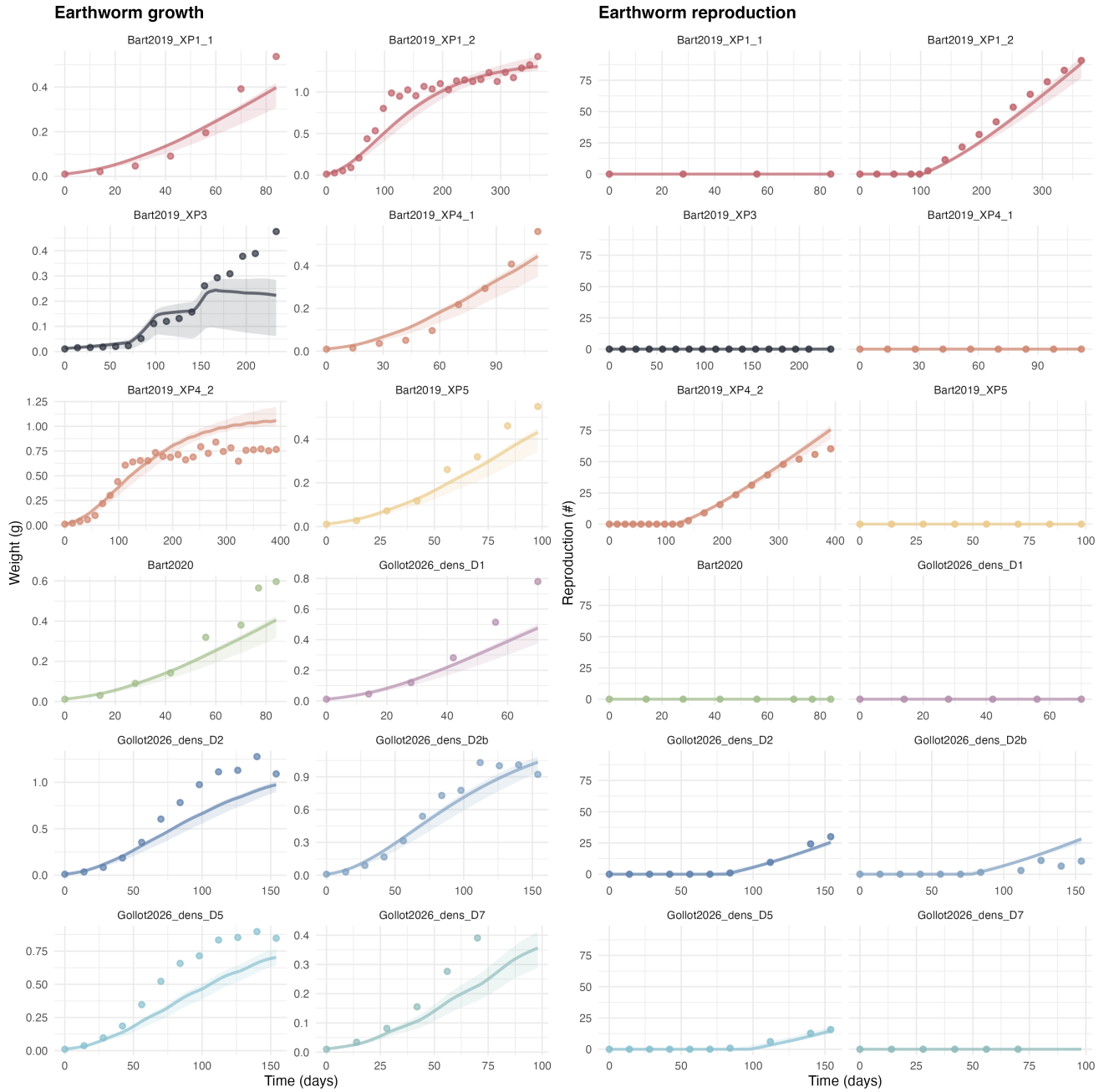


Figure 2.26: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

2.11 Summary

Table 2.2: Estimate summary of each models and their respective log-likelihood and BIC.

Parameter	A_FLYw	B_FLYe	C_CWYe	D_CWKe	E_CLYe	F_CLKe	G_FLKe	H_FWKe
$\{\dot{p}_{Am}\}$	215.8 [207.6 ; 222.6]	129.5 [116.2 ; 155.3]	52.0 [49.9 ; 60.2]	259.0 [237.6 ; 280.0]	135.3 [109.0 ; 157.5]	228.1 [206.0 ; 245.5]	297.3 [268.2 ; 316.8]	302.3 [268.3 ; 329.4]
$[\dot{p}_M]/\{\dot{p}_{Am}\}$ ratio	0.855 [0.786 ; 0.879]	0.655 [0.451 ; 0.697]	0.244 [0.161 ; 0.359]	0.630 [0.548 ; 0.688]	0.593 [0.429 ; 0.695]	0.550 [0.444 ; 0.613]	0.891 [0.849 ; 0.919]	0.917 [0.865 ; 0.949]
$\{\dot{F}_m\}$ or $[\dot{F}_m]$	9.992 [9.85 ; 9.999]	8.571 [7.098 ; 9.907]	9.978 [9.55 ; 9.996]	6.921 [4.076 ; 8.939]	9.007 [6.712 ; 9.919]	6.335 [4.532 ; 7.937]	8.832 [6.803 ; 9.931]	9.666 [5.545 ; 9.956]
r_{OM}	0.052 [0.044 ; 0.061]	0.000 [0 ; 0.01]	0.000 [0 ; 0.047]	0.020 [0.017 ; 0.025]	0.000 [0 ; 0.009]	0.015 [0.012 ; 0.019]	0.017 [0.011 ; 0.025]	0.021 [0.011 ; 0.027]
K	NA	NA	NA	0.078 [0.047 ; 0.108]	NA	0.138 [0.099 ; 0.196]	0.095 [0.069 ; 0.140]	0.058 [0.028 ; 0.074]
v	0.197 [0.168 ; 0.236]	0.181 [0.149 ; 0.221]	0.183 [0.143 ; 0.221]	0.243 [0.208 ; 0.266]	0.191 [0.142 ; 0.217]	0.226 [0.199 ; 0.259]	0.244 [0.210 ; 0.270]	0.252 [0.211 ; 0.273]
κ	0.663 [0.601 ; 0.715]	0.595 [0.522 ; 0.749]	NA	NA	NA	NA	0.983 [0.968 ; 0.985]	0.983 [0.964 ; 0.987]
κ_J	NA	NA	0.644 [0.588 ; 0.690]	0.979 [0.969 ; 0.986]	0.665 [0.556 ; 0.824]	0.974 [0.966 ; 0.986]	NA	NA

Parameter	A_FLYw	B_FLYe	C_CWYe	D_CWKe	E_CLYe	F_CLKe	G_FLKe	H_FWKe
κ_A	NA	NA	0.037 [0.008 ; 0.296]	0.643 [0.563 ; 0.703]	0.698 [0.535 ; 0.808]	0.594 [0.489 ; 0.656]	NA	NA
τ	NA	NA	6.57 [0.81 ; 14.42]	9.66 [5.43 ; 19.19]	6.20 [1.25 ; 14.22]	11.38 [7.72 ; 20.94]	NA	NA
E_h^p	20.08 [20.01 ; 20.92]	60.48 [45.06 ; 73.77]	20.54 [20.04 ; 24.60]	97.62 [60.88 ; 133.57]	58.90 [32.12 ; 75.64]	109.42 [57.66 ; 131.18]	94.81 [78.21 ; 145.66]	93.30 [68.62 ; 147.29]
E_{coc}	1.001 [1.001 ; 1.097]	1.055 [1.009 ; 1.606]	1.023 [1.001 ; 1.103]	68.42 [58.39 ; 75.54]	1.104 [1.006 ; 1.936]	65.34 [57.72 ; 73.80]	3.937 [3.342 ; 7.589]	4.082 [3.161 ; 9.281]
d_V	0.983 [0.903 ; 1.099]	0.993 [0.910 ; 1.097]	1.048 [0.966 ; 1.162]	1.220 [1.147 ; 1.317]	0.976 [0.906 ; 1.103]	1.223 [1.138 ; 1.306]	1.116 [1.042 ; 1.224]	1.119 [1.040 ; 1.223]
d_E	NA	1.2.e-9 [1.3e-9 ; 4.1e-6]	1.9e-9 [1.0e-9 ; 8.6e-7]	1.1e-9 [1.1e-9 ; 5.7e-7]	1.0e-9 [1.2e-9 ; 3.5e-6]	1.1e-9 [1.3e-6 ; 1.4e-6]	1.0e-9 [1.2e-9 ; 1.0e-6]	2.1e-9 [1.2e-9 ; 7.6e-7]
w_E	310.3 [216.5 ; 383.8]	NA	NA	NA	NA	NA	NA	NA
σ_W	0.183 [0.164 ; 0.214]	0.617 [0.548 ; 0.691]	0.610 [0.551 ; 0.698]	0.088 [0.077 ; 0.100]	0.588 [0.545 ; 0.686]	0.078 [0.068 ; 0.089]	0.125 [0.110 ; 0.151]	0.140 [0.124 ; 0.178]
Log-likelihood	-92.01	-251.04	-264.75	23.79	-250.42	52.21	-20.41	-53.61
BIC	240.26	558.33	596.99	25.54	568.33	-31.31	102.69	169.09



Figure 2.27: RMSE for Weight and Reproduction per experiment for each tested model.

3 Isolated adults

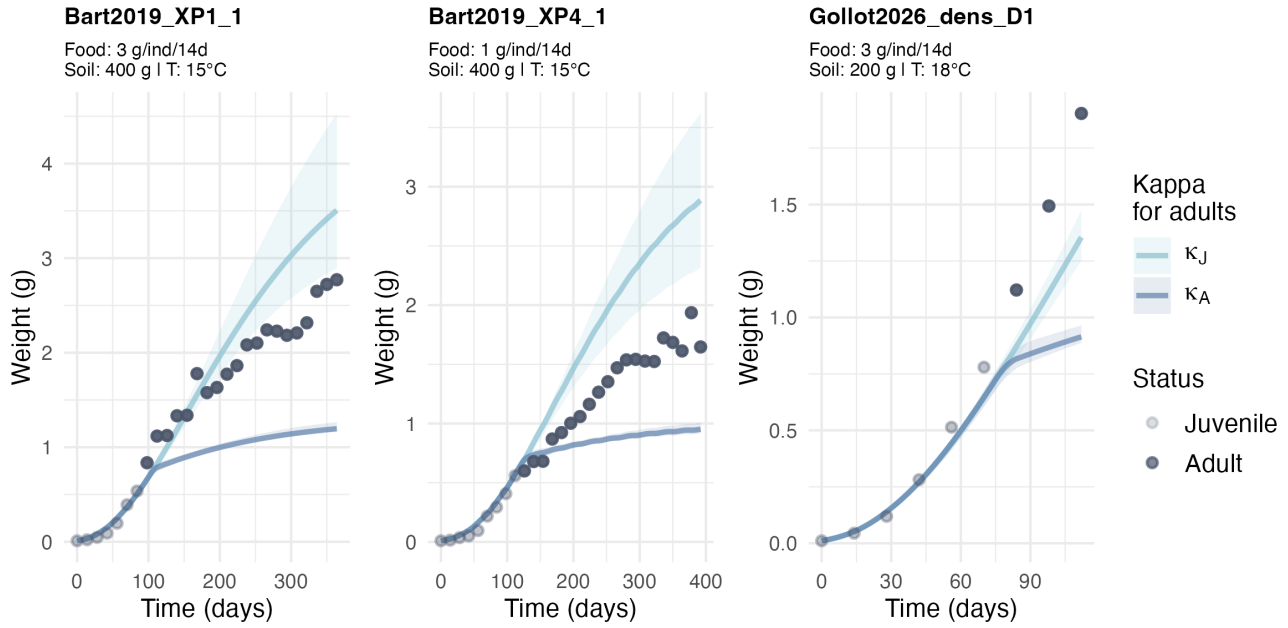


Figure 3.1: Observed and predicted growth trajectories for isolated adults simulated with the retained model (CWKe) under formulations with either a constant juvenile allocation fraction (κ_J , light blue lines) or a transition toward an adult allocation fraction (κ_A , dark blue lines). Solid lines correspond to estimated model trajectories with transparent ribbons corresponding to the 95% credibility intervals. Transparent points represent observations from the juvenile stage, whereas filled points represent observations from the adult stage.

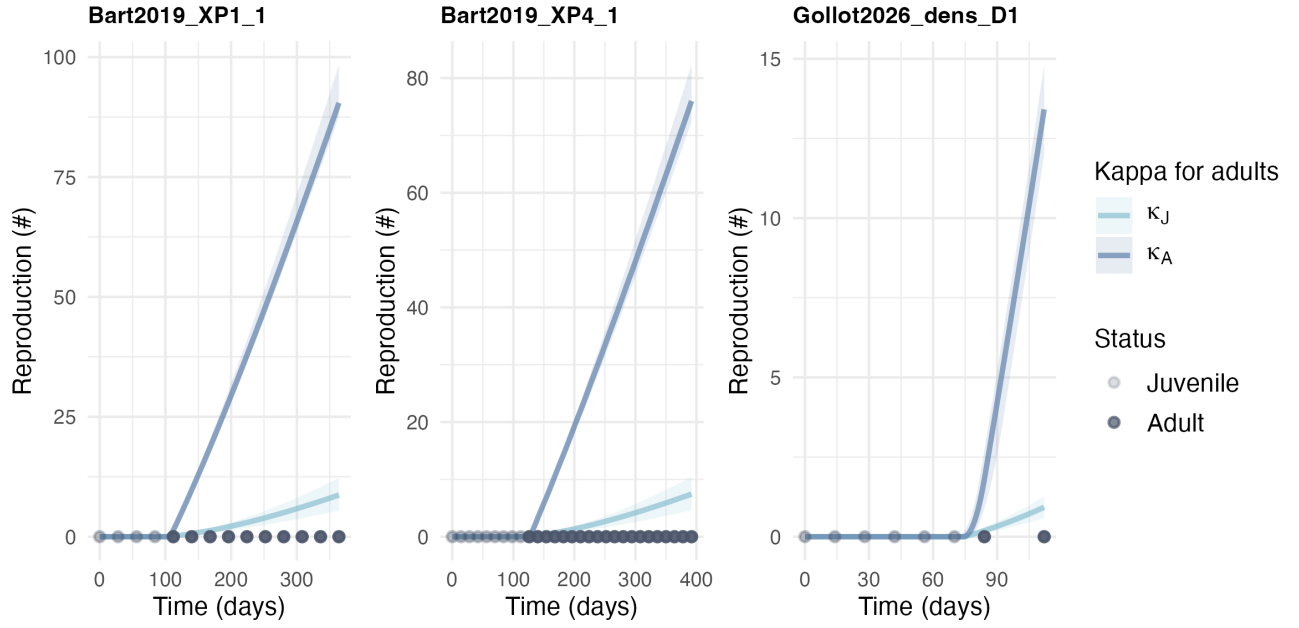


Figure 3.2: Observed and predicted reproduction trajectories for isolated adults simulated with the retained model (CWKe) under formulations with either a constant juvenile allocation fraction (κ_J , light blue lines) or a transition toward an adult allocation fraction (κ_A , dark blue lines). Solid lines correspond to estimated model trajectories with transparent ribbons corresponding to the 95% credibility intervals. Transparent points represent observations from the juvenile stage, whereas filled points represent observations from the adult stage.

4 Trends over environmental conditions

$$\text{Fold} = \begin{cases} Y_{\text{Observed}} \geq Y_{\text{Predicted}}, & \frac{Y_{\text{Observed}}}{Y_{\text{Predicted}}} \\ Y_{\text{Observed}} < Y_{\text{Predicted}}, & \frac{Y_{\text{Predicted}}}{Y_{\text{Observed}}} \end{cases}$$

Table 4.1: Quality of fit for weight and its bias according to experimental temperature.

Experimental temperature (°C)	Mean fold	Mean Log(Pred./Obs.)	exp(Mean Log(Pred./Obs.))
15	1.21	0.10	1.11
18	1.28	-0.08	0.92

Table 4.2: Quality of fit for reproduction and its bias according to experimental temperature.

Experimental temperature (°C)	Mean residual
15	0.0587
18	1.24

Table 4.3: Quality of fit for weight and its bias according to the number of individuals in the cosm and their observed developmental status.

Density (#/cosm)	Status	Mean fold	Mean Log (Pred./Obs.)	exp(Mean Log(Pred./Obs.))
1	Juvenile	1.24	0.13	1.13
2	Juvenile	1.30	0.24	1.27
2	Adult	1.14	-0.08	0.92
5	Juvenile	1.39	-0.33	0.72
5	Adult	1.25	-0.22	0.80
7	Juvenile	1.55	-0.44	0.64
Grouped after puberty	Adult	1.12	0.01	1.01

Table 4.4: Quality of fit for reproduction and its bias according to the number of individuals in the cosm and their observed developmental status.

Density (#/cosm)	Status	Mean residual
1	Juvenile	0.00000
2	Juvenile	0.00000
2	Adult	6.05
5	Juvenile	0.00000
5	Adult	-1.79
7	Juvenile	0.00000
Grouped after puberty	Adult	0.197

Table 4.5: Quality of fit for weight and its bias according to percentage of organic matter (OM) from horse dung given every 14 days per individual and the observed developmental status of earthworms.

OM from horse dung (%/ind/14 d)	Status	Mean fold	Mean Log(Pred./Obs.)	exp(Mean Log(Pred./Obs.))
0.19	Juvenile	1.55	-0.44	0.64
0.22	Juvenile	1.38	0.24	1.34
0.22	Adult	1.13	0.120	1.13
0.27	Juvenile	1.39	-0.33	0.72
0.27	Adult	1.25	-0.22	0.80
0.34	Juvenile	1.09	-0.02	0.98
0.67	Juvenile	1.23	0.10	1.11
0.67	Adult	1.13	-0.12	0.89
1.3	Juvenile	1.28	0.16	1.18
1.3	Adult	1.09	0.003	1.00
0 or 0.67	Juvenile	1.19	0.09	1.10

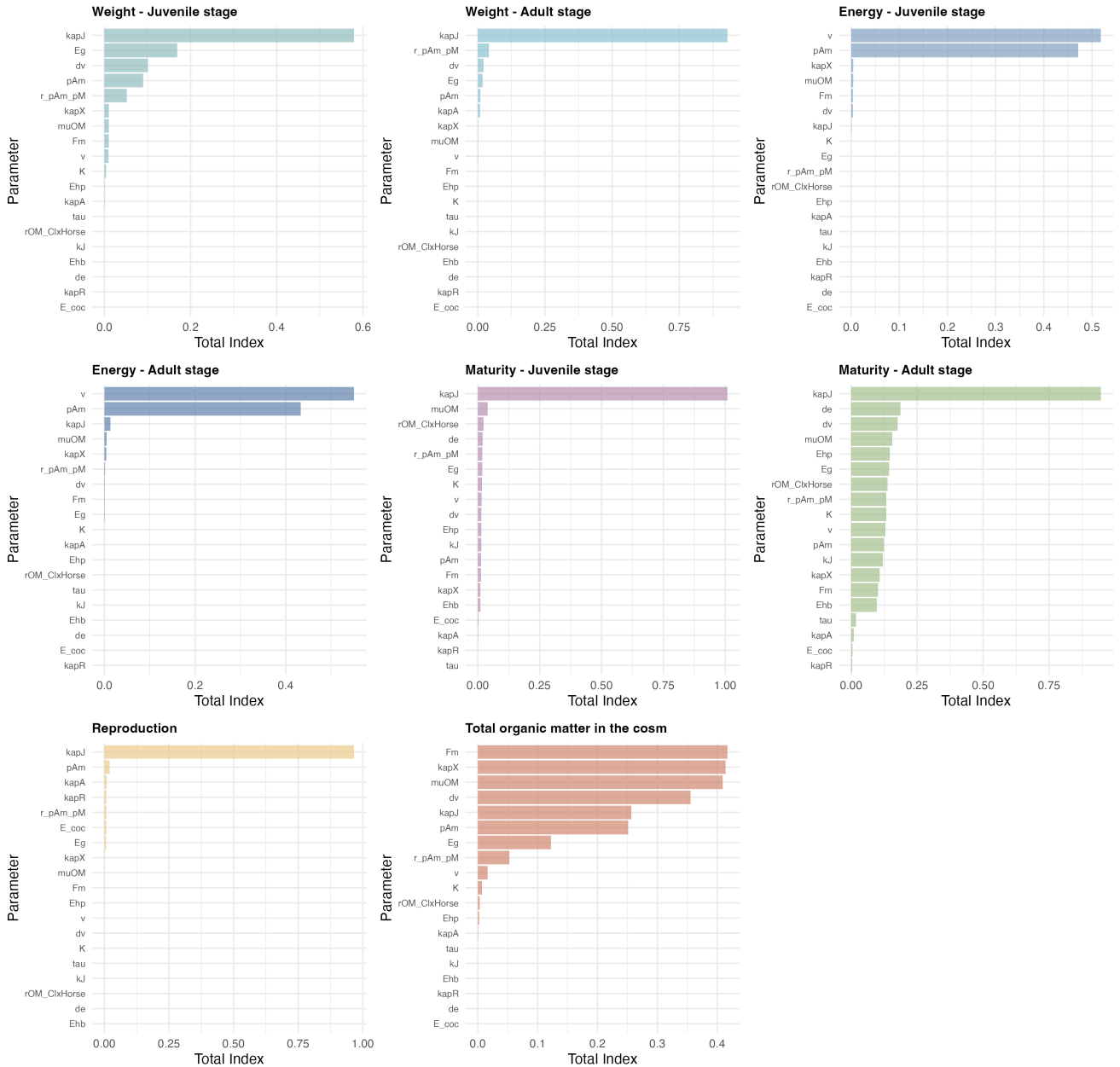
Table 4.6: Quality of fit for reproduction and its bias according to percentage of organic matter (OM) from horse dung given every 14 days per individual and the observed developmental status of earthworms.

OM from horse dung (%/ind/14 d)	Status	Mean residual
0.19	Juvenile	0.000001
0.22	Juvenile	0.000001
0.22	Adult	3.49
0.27	Juvenile	0.000001
0.27	Adult	-1.79
0.34	Juvenile	0.000001

OM from horse dung (%/ind/14 d)	Status	Mean residual
0.67	Juvenile	0.000001
0.67	Adult	-2.50
1.3	Juvenile	0.000001
1.3	Adult	11.6
0 or 0.67	Juvenile	0.000001

5 Sensitivity analysis of best model

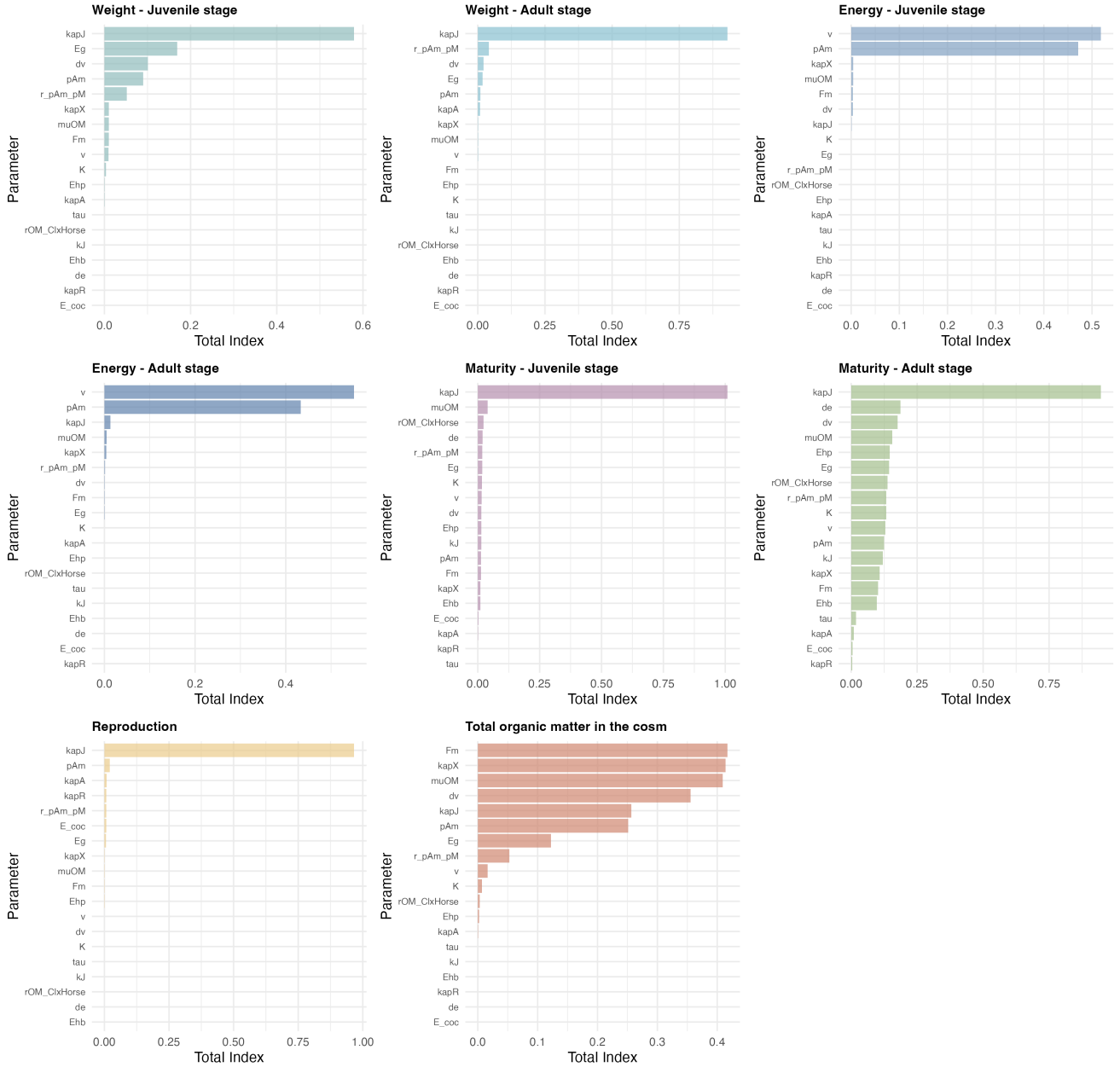
Sobol Sensitivity Analysis - Total Index



Nsim = 30000 | Nboot = 1000

Figure 5.1: **Global Sobol sensitivity analysis of the retained DEB model including temperature-related parameters.** Total-order sensitivity indices are shown for body mass, reserve energy, maturity, reproduction, and total organic matter remaining in the cosm. Endpoints were evaluated at the juvenile stage ($t = 50$ days) and adult stage ($t = 140$ days) using the Gollot2026_D2 simulation scenario (two earthworm per cosm with moderate food). Higher indices indicate a greater contribution of the parameter, including both its direct effect and its interactions with other parameters. 30,000 simulations (Nsim) and 1,000 bootstrap replicates (Nboot) were used.

Sobol Sensitivity Analysis - Total Index



Nsim = 30000 | Nboot = 1000

Figure 5.2: **Global Sobol sensitivity analysis of the retained DEB model excluding temperature-related parameters.** Total-order sensitivity indices are shown for body mass, reserve energy, maturity, reproduction, and total organic matter remaining in the cosm. Endpoints were evaluated at the juvenile stage ($t = 50$ days) and adult stage ($t = 140$ days) using the Gollot2026_D2 simulation scenario (two earthworm per cosm with moderate food). Excluding thermal parameters allowed the relative influence of physiological, ingestion-related, and allocation parameters to be examined independently of temperature correction. 30,000 simulations (Nsim) and 1,000 bootstrap replicates (Nboot) were used.

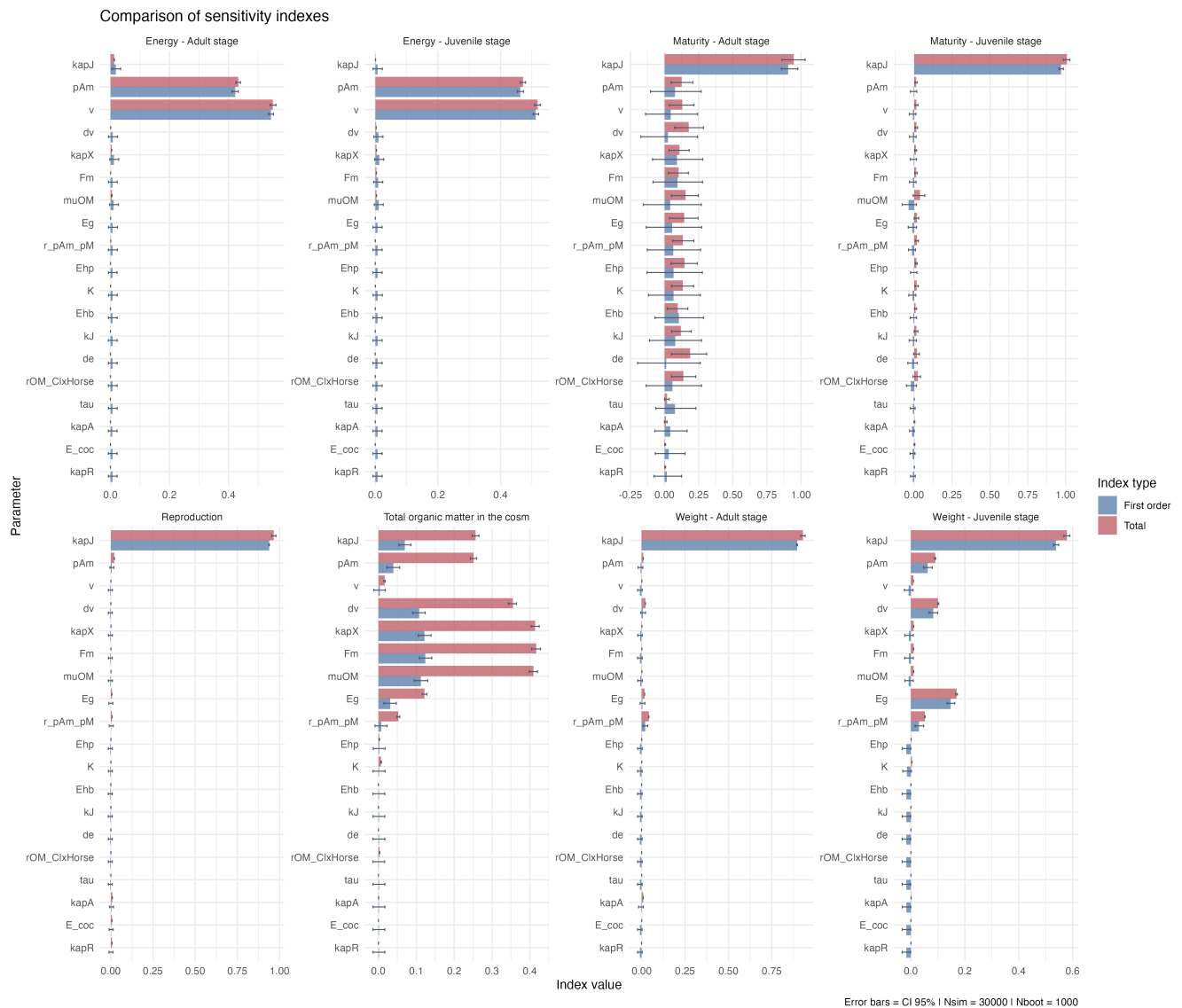


Figure 5.3: **Comparison of first-order and total-order Sobol sensitivity indices for the retained DEB model.** First-order indices represent the direct contribution of each parameter to output variance, whereas total-order indices include both direct effects and interactions with other parameters. Error bars represent 95% confidence intervals obtained from 1,000 bootstrap replicates (Nboot) and 30,000 simulations (Nsim).

- Bart, S., Laurent, C., Péry, A. R. R., Mougin, C., & Pelosi, C. (2017). Differences in sensitivity between earthworms and enchytraeids exposed to two commercial fungicides. *Ecotoxicology and Environmental Safety*, 140, 177–184. <https://doi.org/10.1016/j.ecoenv.2017.02.052>
- Cummins, K. W., & Wuycheck, J. C. (1971). Caloric Equivalents for Investigations in Ecological Energetics. *SIL Communications, 1953-1996*, 18(1), 1–158. <https://doi.org/10.1080/05384680.1971.11903918>
- Driver, E. A., Sugden, L. G., & Kovach, R. J. (1974). Calorific, chemical and physical values of potential duck foods. *Freshwater Biology*, 4(3), 281–292. <https://doi.org/10.1111/j.1365-2427.1974.tb00098.x>
- Eggleton, M. A., & Schramm, H. L. (2004). Feeding Ecology and Energetic Relationships with Habitat

of Blue Catfish, *Ictalurus furcatus*, and Flathead Catfish, *Pylodictis olivaris*, in the Lower Mississippi River, U.S.A. *Environmental Biology of Fishes*, 70(2), 107–121. <https://doi.org/10.1023/B:EBFI.0000029341.45030.94>

Kabus, J., Hartmann, V., Cocchiararo, B., Dombrowski, A., Enns, D., Karaouzas, I., Lipkowski, K., Pelikan, L., Shumka, S., Soose, L., Baker, N. J., & Jourdan, J. (2024). Cryptic species complex shows population-dependent, rather than lineage-dependent tolerance to a neonicotinoid. *Environmental Pollution*, 362, 124888. <https://doi.org/10.1016/j.envpol.2024.124888>